MEDIPOST

카티스템®
제품 설명자료

메디포스트㈜ 카티스템사업본부 학술마케팅팀

김경식

Life-Changing **INNOVATIONS**

오늘의 한계를 뛰어넘는 내일의 혁신으로 생명공학 분야의 새 역사를 만들어 갑니다.

93건	38건	334,205units	45%
국내외 특허건수	국책과제 수행	줄기세포은행 보관건수	연구개발인력

(2026년 3월 기준)

- **회사명** : 메디포스트㈜ (MEDIPOST CO., Ltd.)
- **대표이사** : 오원일
- **설립일자** : 2000년 6월 26일
- **상장일자** : 2005년 7월 29일 (078160KQ)
- **임직원수** : 346명
- **본사 및 법인**
  - 한국 본사 : 성남시 분당구 (판교테크노밸리)
  - 생산 공장 : 서울시 구로구
  - 미국 법인 : 미국 델라웨어주
- **주요성과**
  - 세계 최초 동종 제대혈 줄기세포치료제 출시(카티스템®)
  - 연구개발 파이프라인 : 무릎연골재생, 폐질환
  - 줄기세포은행 누적 보관 및 이식 1위
- **주요 사업 영역 소개**

- **줄기세포치료제 카티스템®** : 제대혈유래 줄기세포 신약 개발
- **CDMO** : 세포치료제 위탁 개발 생산
- **제대혈은행** : 대한민국 대표 제대혈은행, 셀트리
- **건강기능식품** : 가족 맞춤 영양 솔루션, 모비타

VISION

세계 1위 줄기세포 기술의
글로벌 생명공학 리더

CORE VALUE

인재 중심 : 도전, 혁신, 열정
고객 중심 : 생명존중,
가치 창조, 신뢰

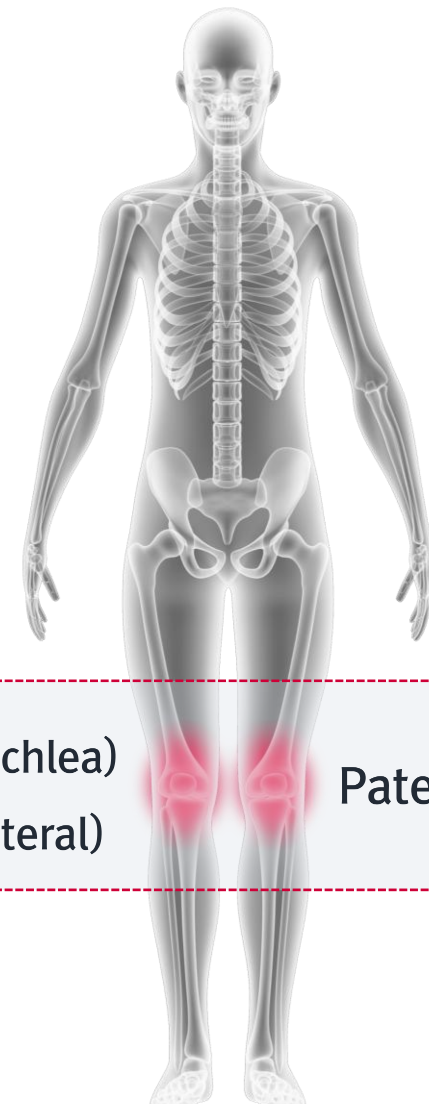
MISSION

건강과 생명 연장을 위한
토털 솔루션 제공
인류의 삶의 질 향상에 기여

세계 최초 동종 제대혈유래 줄기세포치료제 카티스템®

카티스템® 제품 개요

- 적응증 : 퇴행성 관절염 등으로 손상된(ICRS Grade IV) 무릎연골재생
- 식약처 품목허가('12.1/전문의약품, 비급여)



Distal Femur (Medial / Lateral / Trochlea)

Tibia (Medial / Lateral)

Patellar

제품 포장 정보

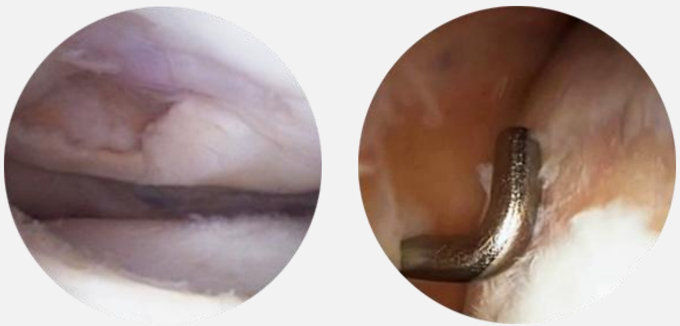
CARTISTEM® 제품 포장 정보

바이알	구성성분	기능	함량*(바이알당)	저장방법 및 사용기간
1	동종 제대혈유래 중간엽줄기세포	주성분	7,500,000cells	저장방법 : 밀봉용기/4~20°C 사용기간 : 제조일로부터 48시간 (밀봉완료 된 제조시로부터 48시간)
	알파-MEM (페놀레드 무첨가)	현탁액	1.5 mL	
2	히알루론산나트륨	첨부물	60 mg	

세계 최초 동종 제대혈유래 줄기세포치료제 카티스템®

카티스템® 개요

시술 전

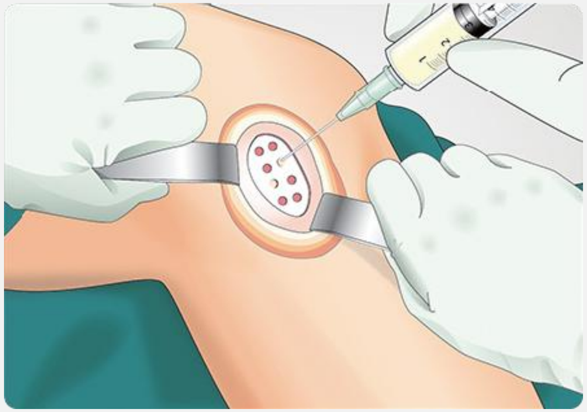


카티스템®

[주성분] **동종제대혈유래중간엽줄기세포**
- 병변 크기에 따른 유효 용량(세포수) 적용
- 손상된 연골조직의 재생 도모
- 정형외과적 수술이 가능한 모든 연령 적용

[첨부물] **히알루론산나트륨**
- 동결건조된 생체적합성 고분자
- 세포와의 혼합 시 세포적합성 확보
- 혼합 후 하이드로겔 메트리스 형태로 병변의 다양한 크기 및 모양에 적용

카티스템 수술



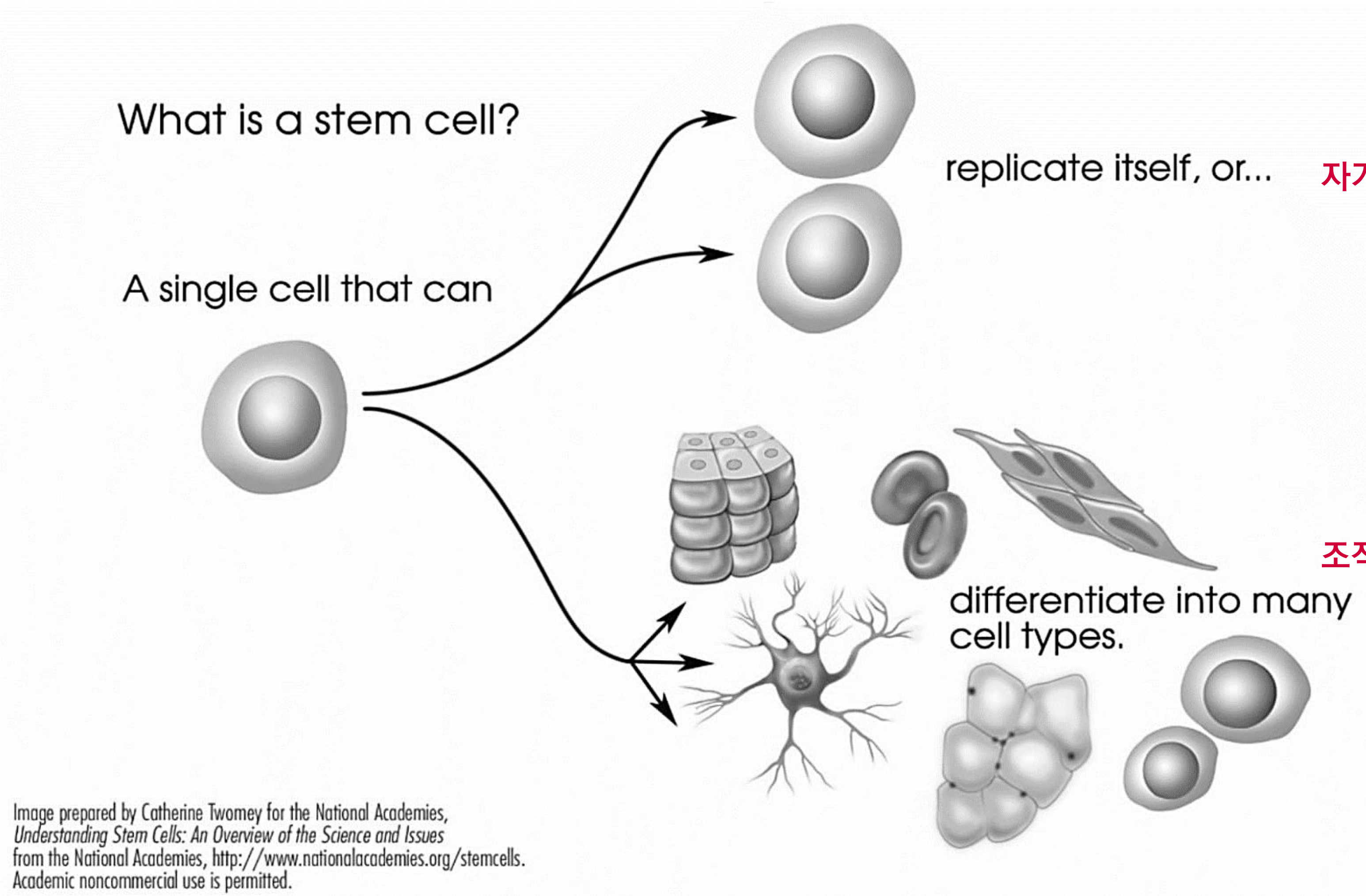
1년 후 관절연골 재생

줄기세포란(stem cell)?

CARTISTEM

자가복제능력을 가지고 있으며 우리 몸을 구성하는 조직세포로 분화할 수 있는

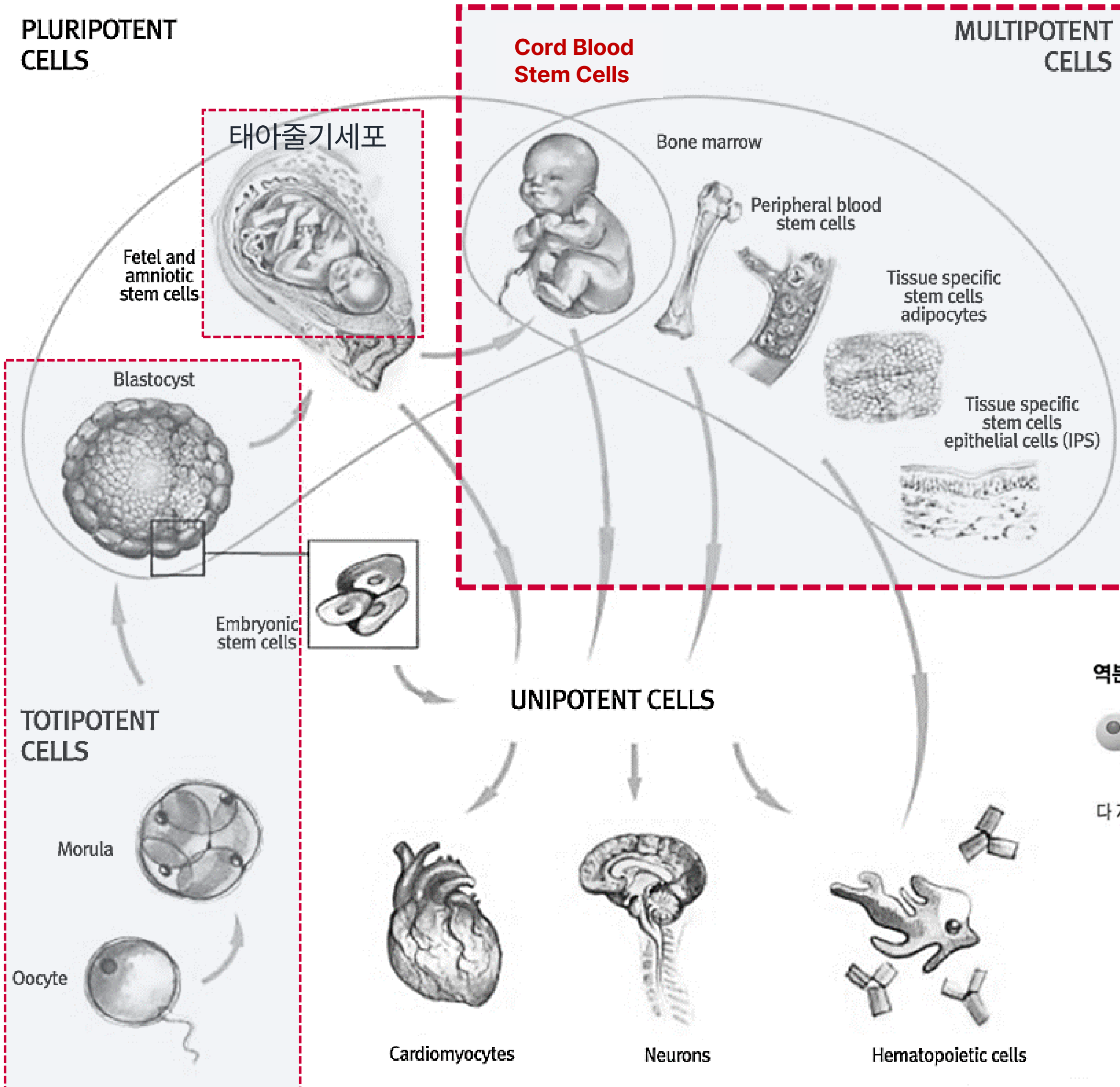
'미분화 상태의 만능세포'



자가복제능력

조직세포 분화능

줄기세포의 종류



성체줄기세포

인체 일부 조직으로만 분화 가능

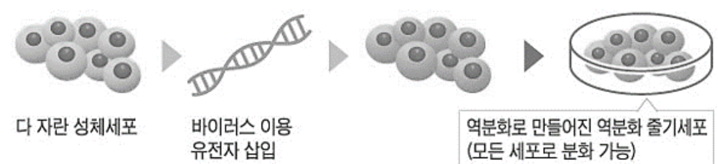
조혈줄기세포 (Hematopoietic Stem Cell; HSC)

➔혈액 세포(적혈구, 백혈구 등)으로 분화됨.

중간엽줄기세포 (Mesenchymal Stem Cell; MSC)

➔연골, 뼈, 근육, 인대, 지방 으로 분화됨.

역분화 줄기세포



배아줄기세포

인체 다양한 조직으로 분화 가능

The Future of Biotechnology, MEDIPOST

중간엽줄기세포(Mesenchymal Stem Cells, MSC) 정의

CARTISTEM

- 부착성 세포(Adherence to Plastic)
- 특이세포표면항원 발현(Specific Cell Surface Antigen (Ag) Expression)
- 다능성 분화능(Multipotent Differentiation Potential)

by ISCT(International Society for Cellular Therapy)

- 부착성 세포군

➔ 형태학적, 생물학적 변화 없이 대량 생산 및 저장 가능

카티스템® GMP 생산 과정

제대혈 채취

공여자 적격성 검사

세포 분리

반복 계대배양

세포 동결 저장

· -196°C

제품 주문

· 매주 화, 목요일

세포 해동 및 배양

·8일 소요
·품질관리
·유효기간(48시간: 수,금 18:00까지)

카티스템®
[주성분] 동종제대혈유래중간엽줄기세포
[첨부물] 동결건조된 히알루론산나트륨

발주	월	화(주문)	수	목(주문)	금
수술	월	화(수술)	수(수술)	목(수술)	금(수술)

중간엽줄기세포 정의

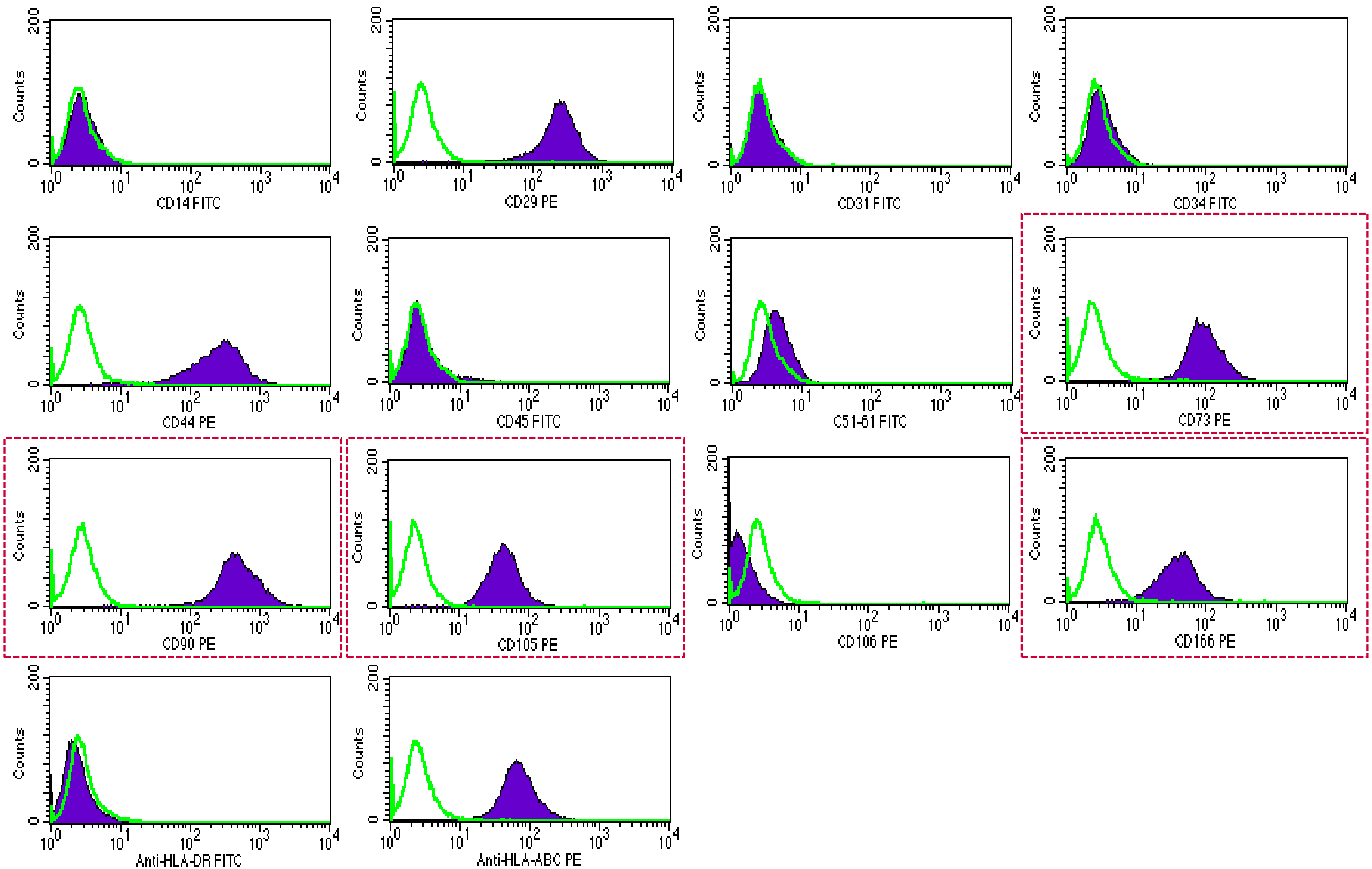
CARTISTEM

카티스템® 품질검사 항목

시험항목		판정기준	
성상시험		무색투명한 바이알에 연황색의 세포가 무색의 현탁액에 부유, 침전된 액상 주사제	
무균시험(MFM)	3일차	균의 발육이 인정되지 않아야 함	
	최종	균의 발육이 인정되지 않아야 함(약 10일 소요)	
마이코플라스마 부정시험 (Real-Time PCR)		마이코플라스마의 증식이 인정되지 않아야 함	
엔도톡신 시험		〈 2 EU/mL	
총 세포수 측정시험		총 세포수 6.8 ~ 8.3 x 10^6 cells/vial	
세포 생존율 시험		≥80 %	
확인시험 (줄기세포 특이세포표현형 확인시험)		CD73	≥ 80.0%
		CD90	≥ 85.0%
		CD105	≥ 85.0%
		CD166	≥ 80.0%
순도시험		CD14	≤ 1.0%
		CD45	≤ 1.0%
		HLA-DR	≤ 1.0%
역가시험 (단백질 발현량 확인 시험)		≥ 1,613 pg/mL	

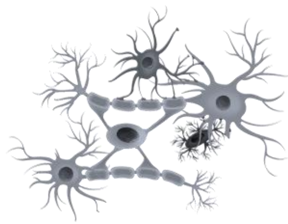
중간엽줄기세포 정의

특이세포표면항원 발현



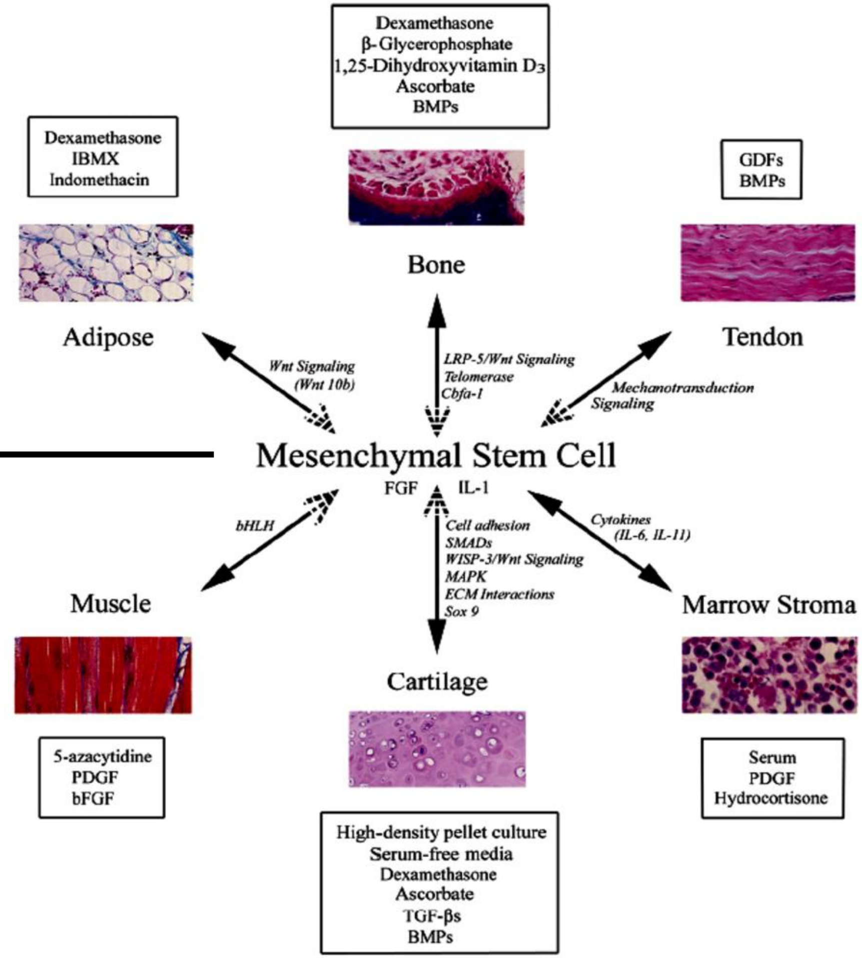
다능성 분화능

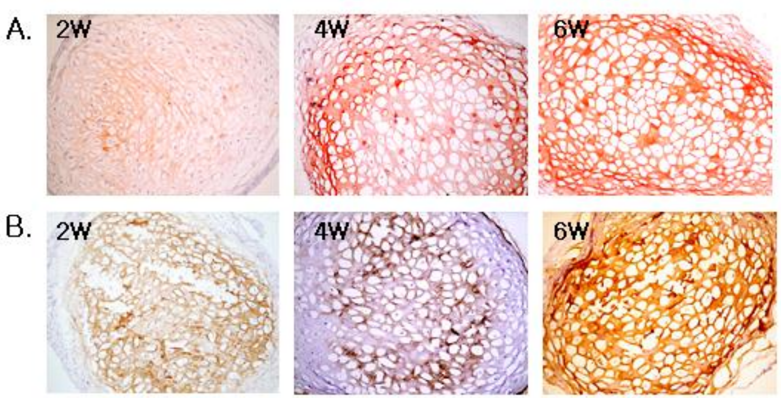
Multipotent somatic stem cells



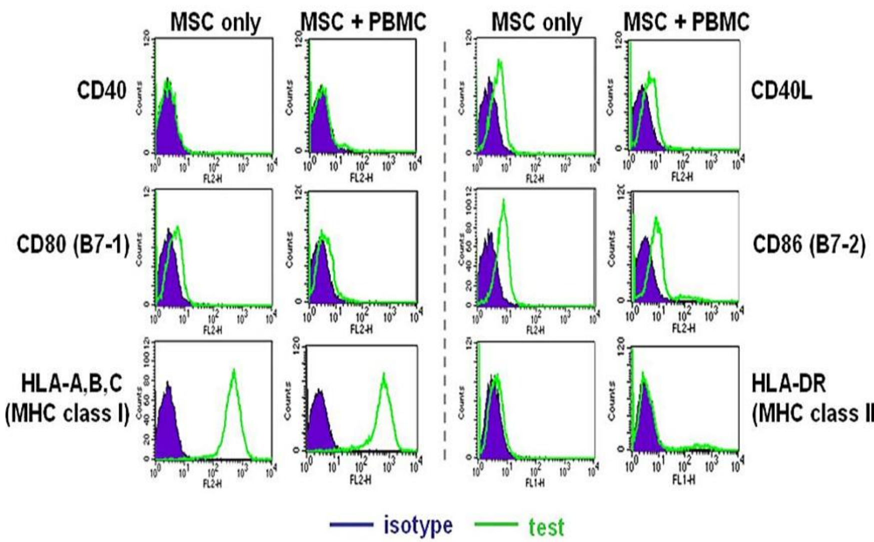
Neuron

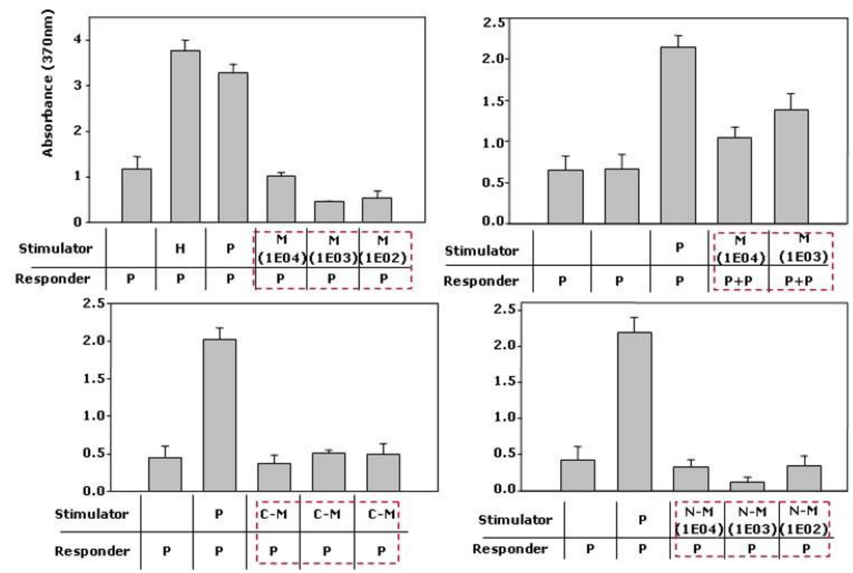
Isobutylmethylxanthine
dibutyryl cyclic AMP
EGF, BDNF
5-azacytidine
neurotrophin-3
DMSO/butylated
hydroxyanisole





면역억제 및 조절능

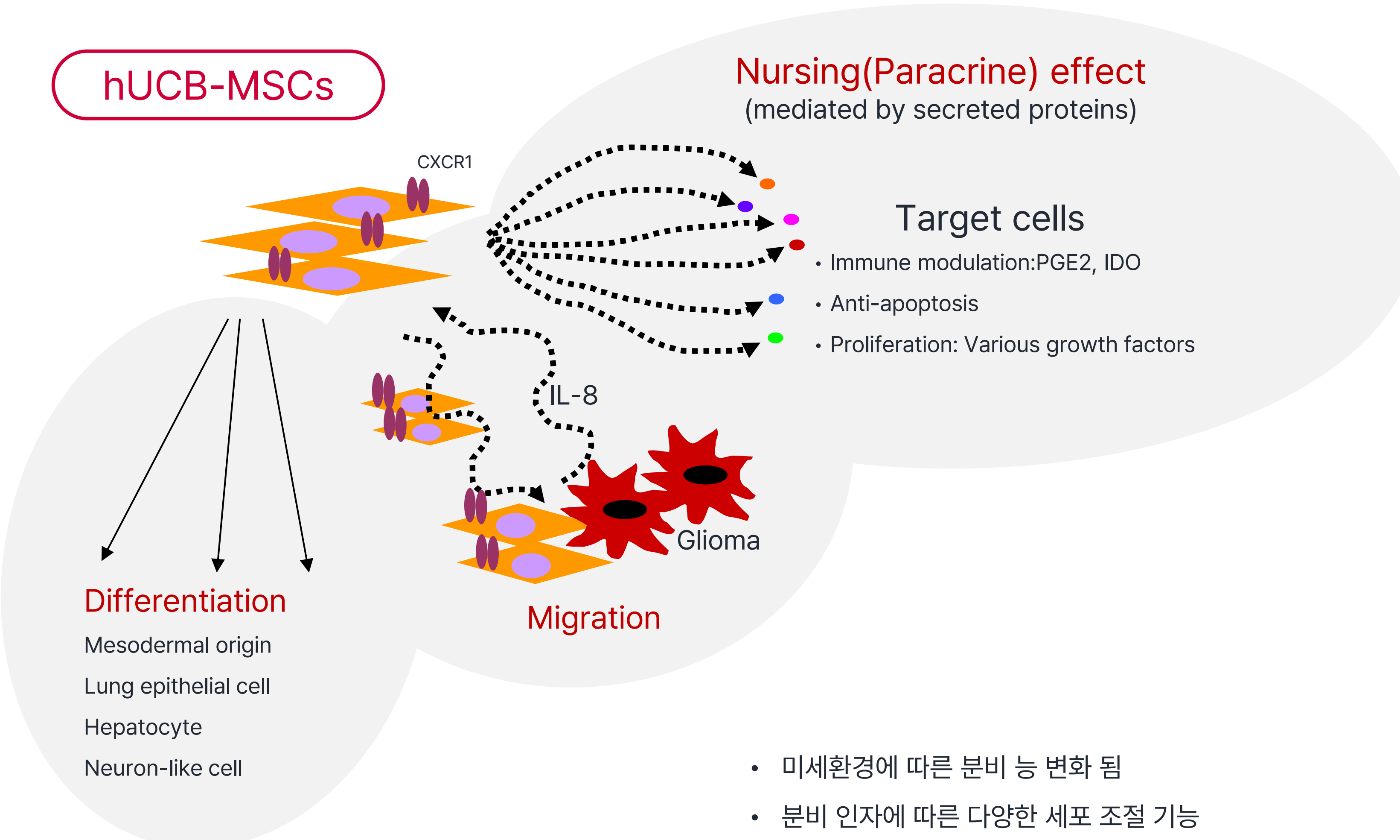




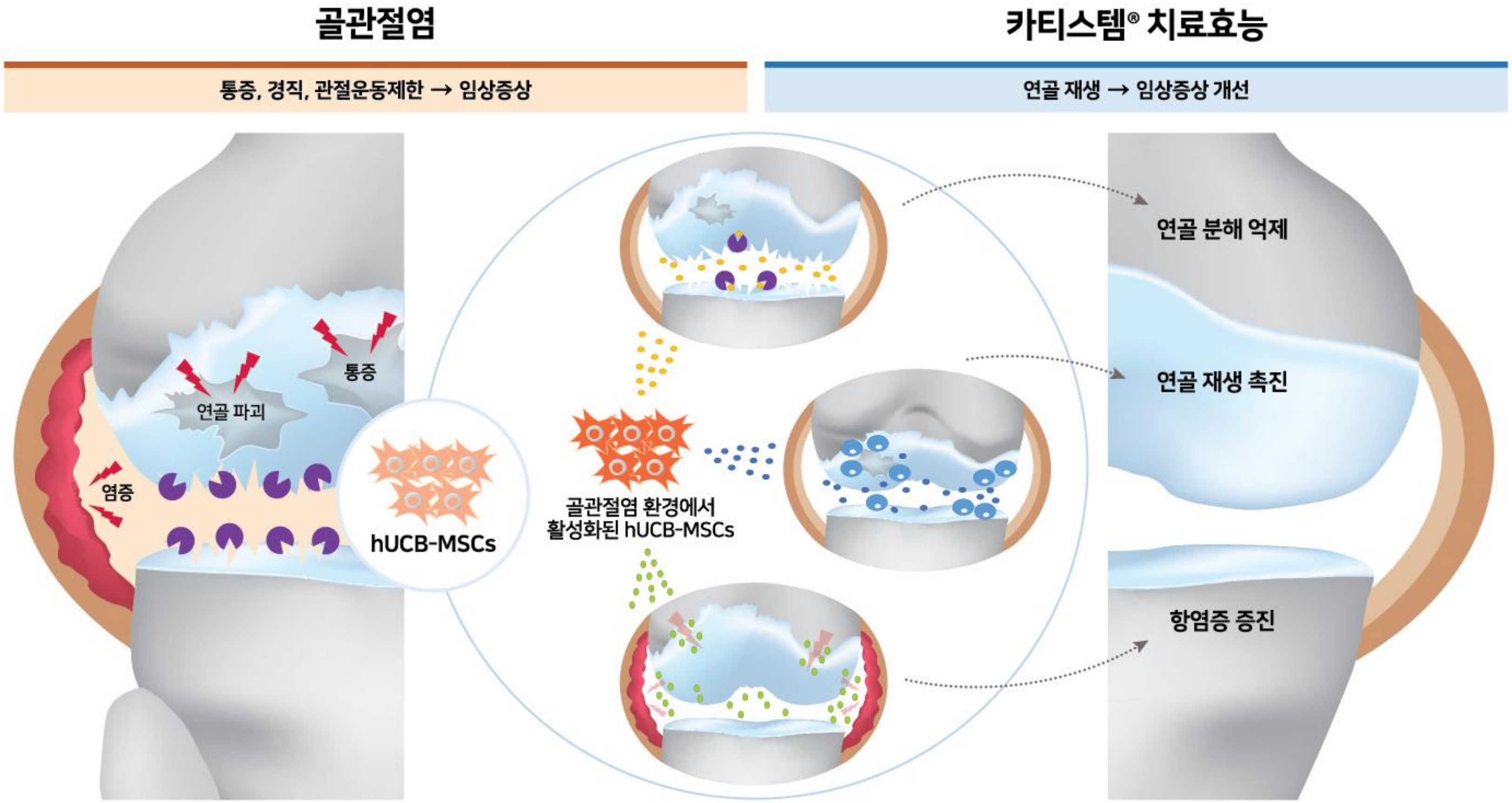
- 세포 표면에 면역 활성 인자 발현이 없음
- 면역활성 상황 및 분화 전과 후에서도 면역적 안전성을 나타냄
- 면역 조절능을 가짐 (T림프구 증식 억제)

➔ 동종 이식에 적합

Paracrine effect



- 미세환경에 따른 분비 능 변화 됨
- 분비 인자에 따른 다양한 세포 조절 기능
  (항 염증, 세포사멸억제, 세포 증식 촉진, 세포 분화의 유도)

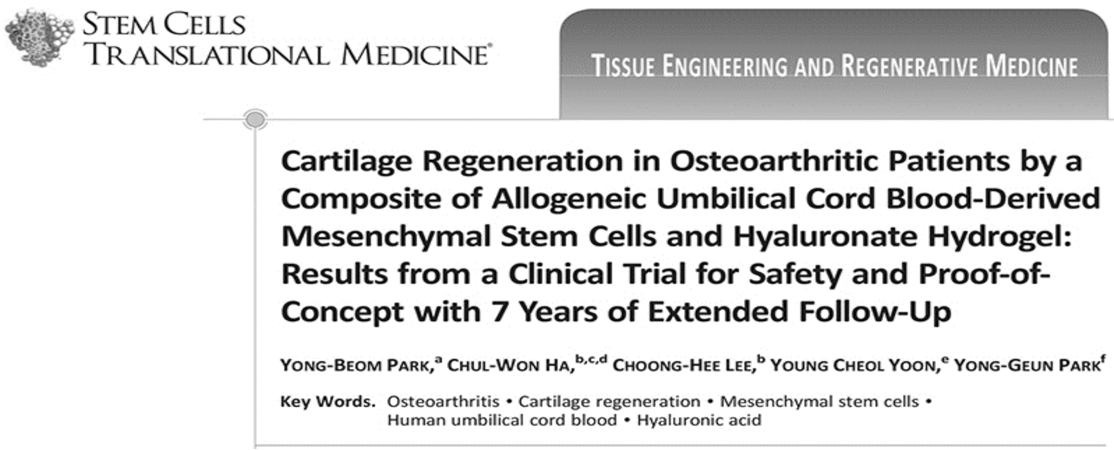


연골기질분해 효소	연골세포
염증	효소 활성 억제 인자
hUCB-MSCs	항염증 인자
골관절염 환경에서 활성화된 hUCB-MSCs	연골분화 촉진 인자

Research results

- 연골 분해 억제 : To be published
- 연골 재생 촉진 : Jeong et al(2013) Stem Cells. 31(10):2136-2148, Park et al(2015) J Orthop Res. 33(11):1580-1586, Ha et al(2015) Stem Cells Transl Med 4(9):1044-1051, Park et al(2016) PLoS One 11(11):e0165446
- 항염증 증진 : To be published

카티스템® 임상 1/2a의 7년차 결과



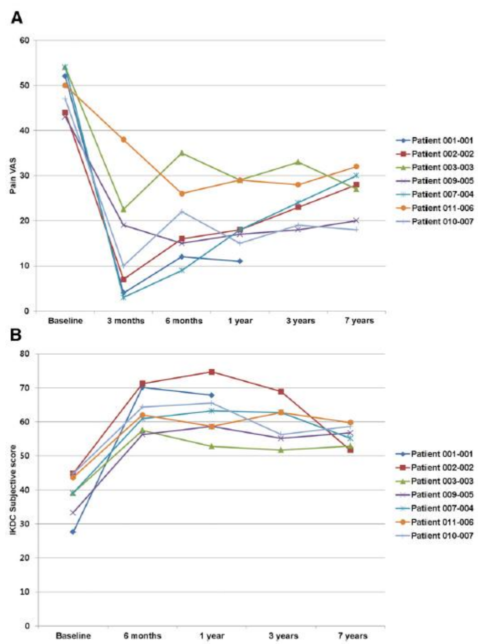
Stem Cells Translational Medicine

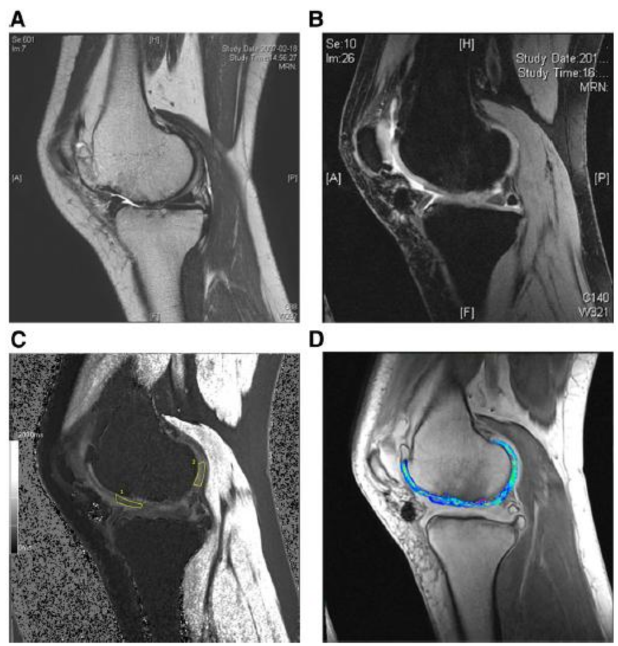
Tissue Engineering and Regenerative Medicine

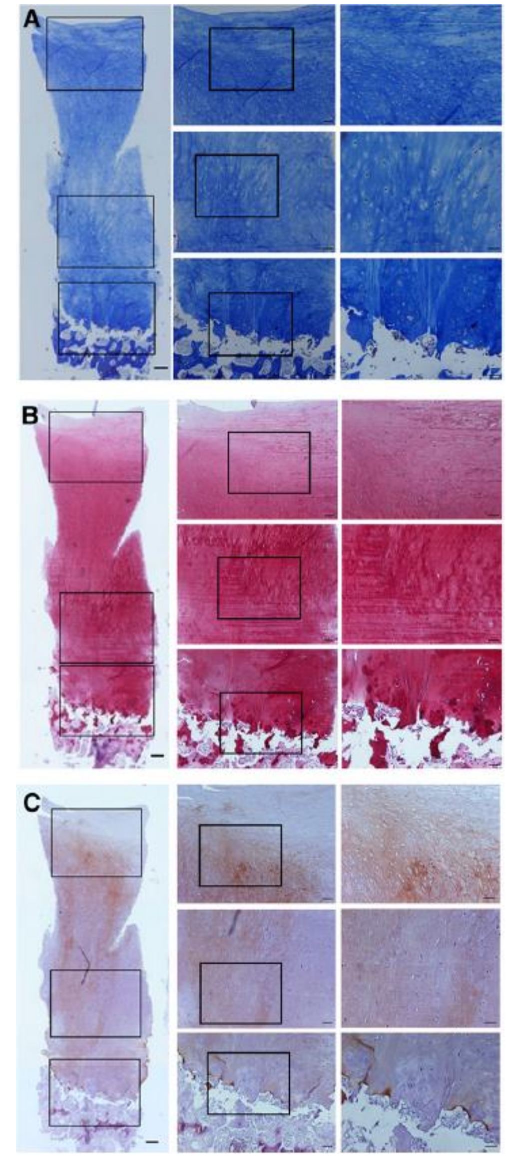
Cartilage Regeneration in Osteoarthritic Patients by a Composite of Allogeneic Umbilical Cord Blood-Derived Mesenchymal Stem Cells and Hyaluronate Hydrogel: Results from a Clinical Trial for Safety and Proof-of-Concept with 7 Years of Extended Follow-Up

Yong-Beom Park,[a] Chul-Won Ha,[b,c,d] Choong-Hee Lee,[b] Young Cheol Yoon,[e] Yong-Geun Park[f]

Key Words. Osteoarthritis • Cartilage regeneration • Mesenchymal stem cells • Human umbilical cord blood • Hyaluronic acid

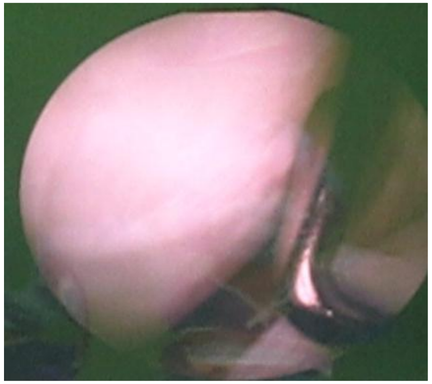




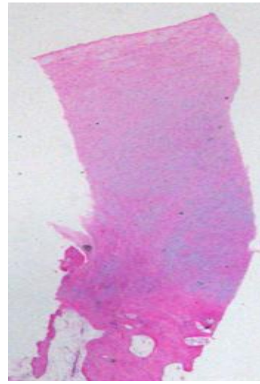


Biopsy (Histologic findings)

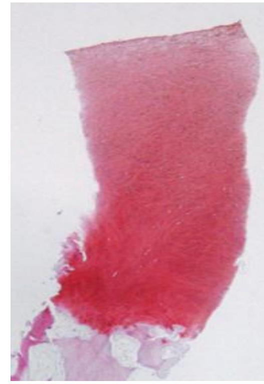
Test Group; CARTISTEM®



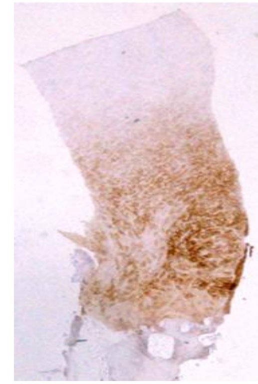
65/F, Grade IV (2.25 cm^2)



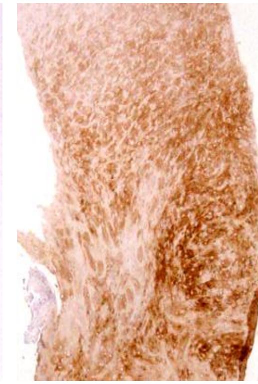
H&E(x20)



Safranin O (x20)



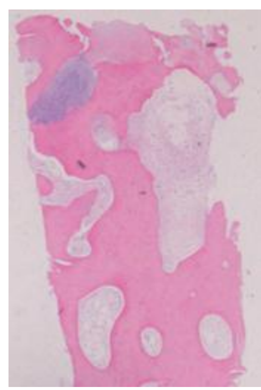
Collagen II IHC(x20)



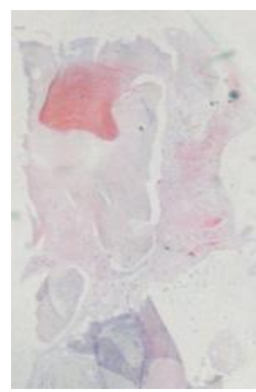
(x40)

Control Group; Microfracture

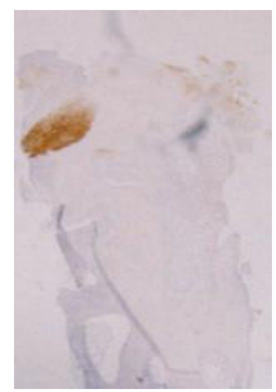
28/M, Grade IV (2.25 cm^2)



H&E(x20)



Safranin O (x20)



Collagen II IHC(x20)

(x40)

카티스템® 3상 임상시험

CARTISTEM

카티스템® 임상시험

무릎연골손상 혹은 결손 환자를 대상으로 한 동종제대혈유래간엽줄기세포 제제(카티스템®)치료군과 대조군(Microfracture)간의 유효성 및 안전성 평가 비교를 위한 무작위배정, 공개, 다기관, 제 3상 임상시험

카티스템® 임상시험 실시기관

임상시험실시기관	시험책임자
가천의대 길병원	이범구
강남 세브란스병원	최종혁
강북 삼성병원	정화재
고려대학교 구로병원	임홍철*
삼성서울병원	하철원
서울보훈병원	윤정로
서울아산병원	빈성일
이화여자대학교 의과대학 부속 목동병원	유재두
인하대병원	김명구
한양대학교병원	최충혁

* Coordinating Investigator

카티스템® 적응증1

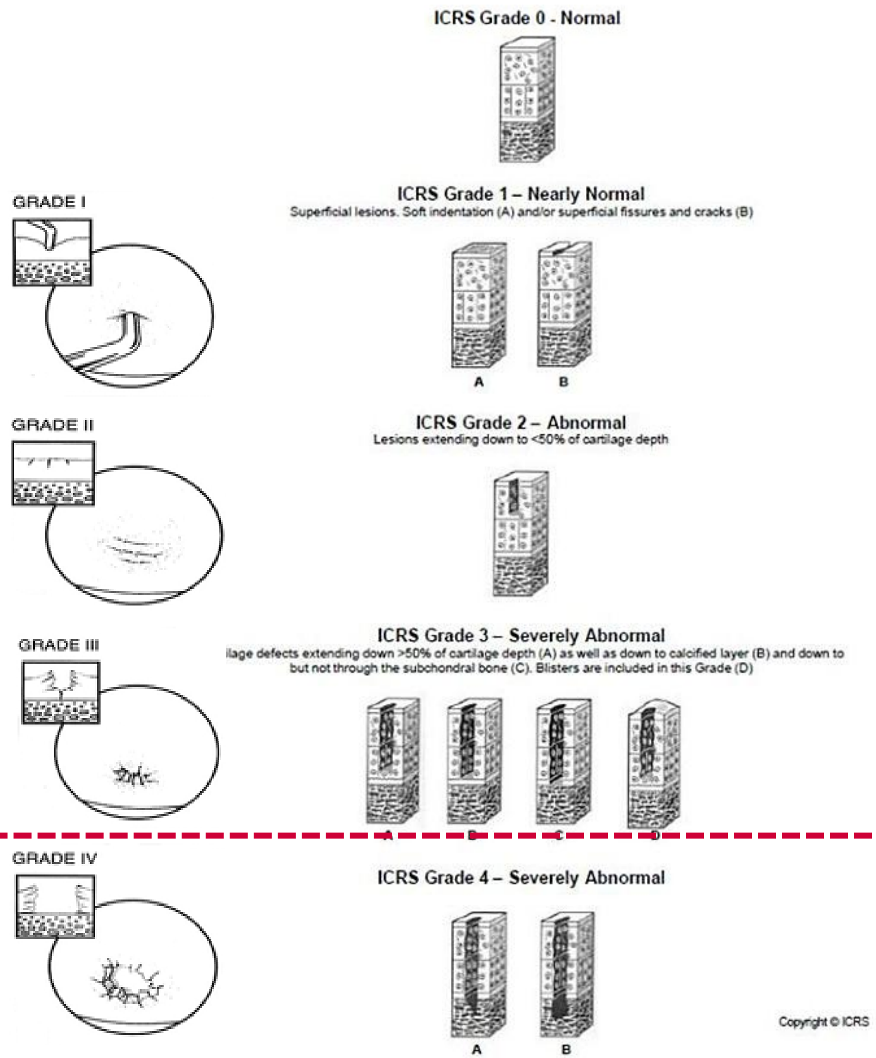
OA, OCD,

ICRS Grade IV,

K&L Grade I~III,

2.5X10^{6}Cells/500ul/cm^2,

No Age Limit



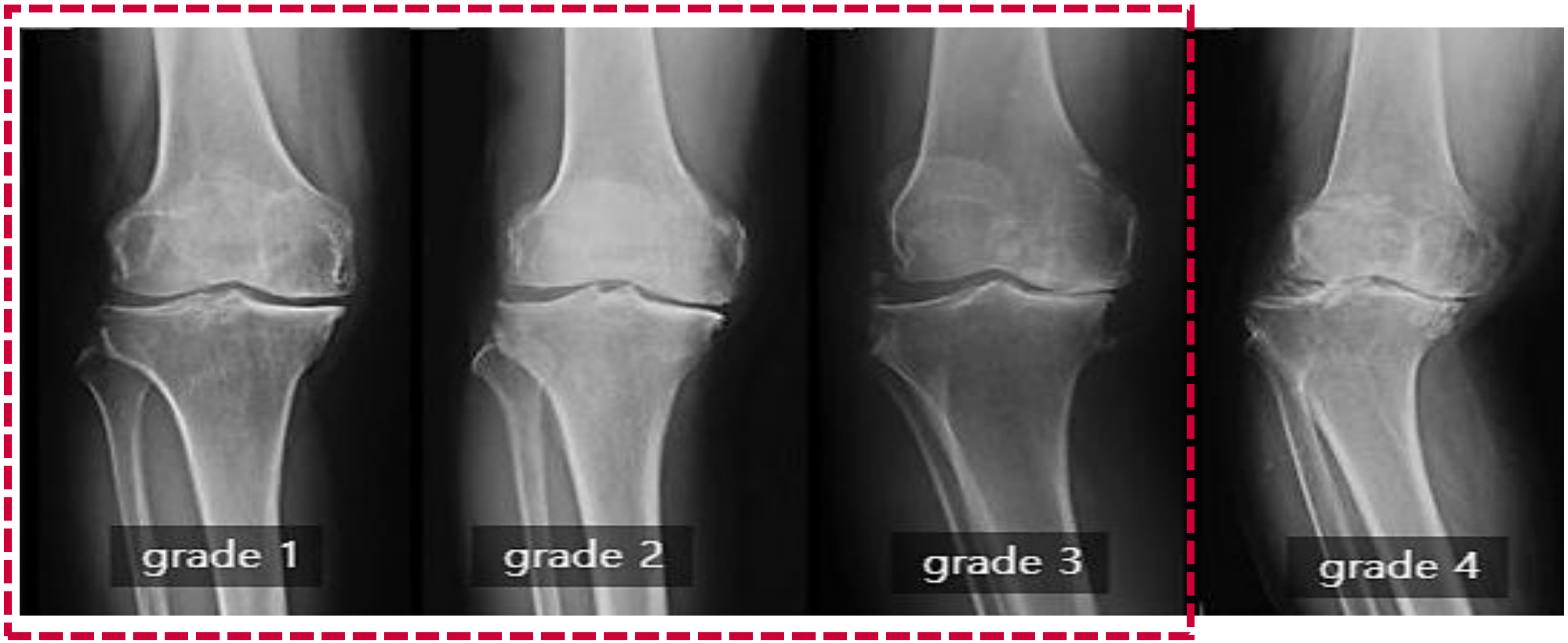
OA, OCD,

ICRS Grade IV,

K&L Grade I~III,

2.5X10^6Cells/500ul/cm^2,

No Age Limit



카티스템® 용량, 용법

병변에 따라 $2.5X10^6$Cells/500ul/cm^2 적용하는 제품

결손면적	키트* 구성용량	투여양
2㎠ 이상~**3㎠ 미만**	각 1 바이알	1.0mL 이상~1.5mL 미만
3㎠ 이상~4㎠ 미만	각 2 바이알	1.5mL 이상~2.0mL 미만
4㎠ 이상~5㎠ 미만		2.0mL 이상~2.5mL 미만
5㎠ 이상~**6㎠ 미만**		2.5mL 이상~3.0mL 미만
6㎠ 이상~7㎠ 미만	각 3 바이알	3.0mL 이상~3.5mL 미만
7㎠ 이상~8㎠ 미만		3.5mL 이상~4.0mL 미만
8㎠ 이상~9㎠ 미만		4.0mL 이상~4.5mL 미만
9 ㎠		4.5mL

카티스템®(동종 제대혈유래 중간엽줄기세포, 히알루론산나트륨)

단위 병변 당 적정 용량 사용으로 효능 극대화!

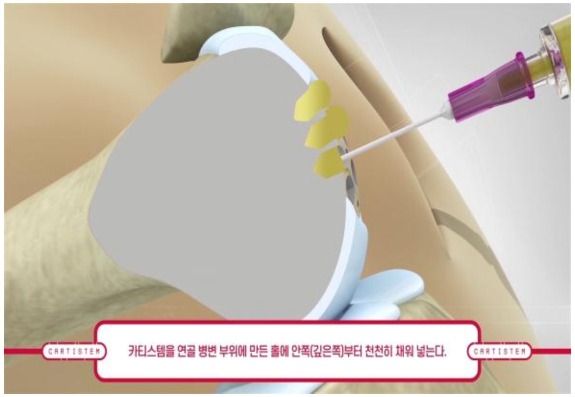
카티스템® 수술 방법

제조시로부터
48시간 내 사용
CARTISTEM
주성분 : 제대혈유래중간엽줄기세포 / 첨부물 : 동결건조된 히알루론산나트륨

5ml 무균주사기로 옮겨진 세포 부유액을 HA에 직접 점적한다.
(점적 전 needle 교체)

관절 연골 병변 부위에 관절강 최소 절개로 노출한 후,
연골 병변 부위의 불완전한 연골(cartilage flap)을 정리한다.

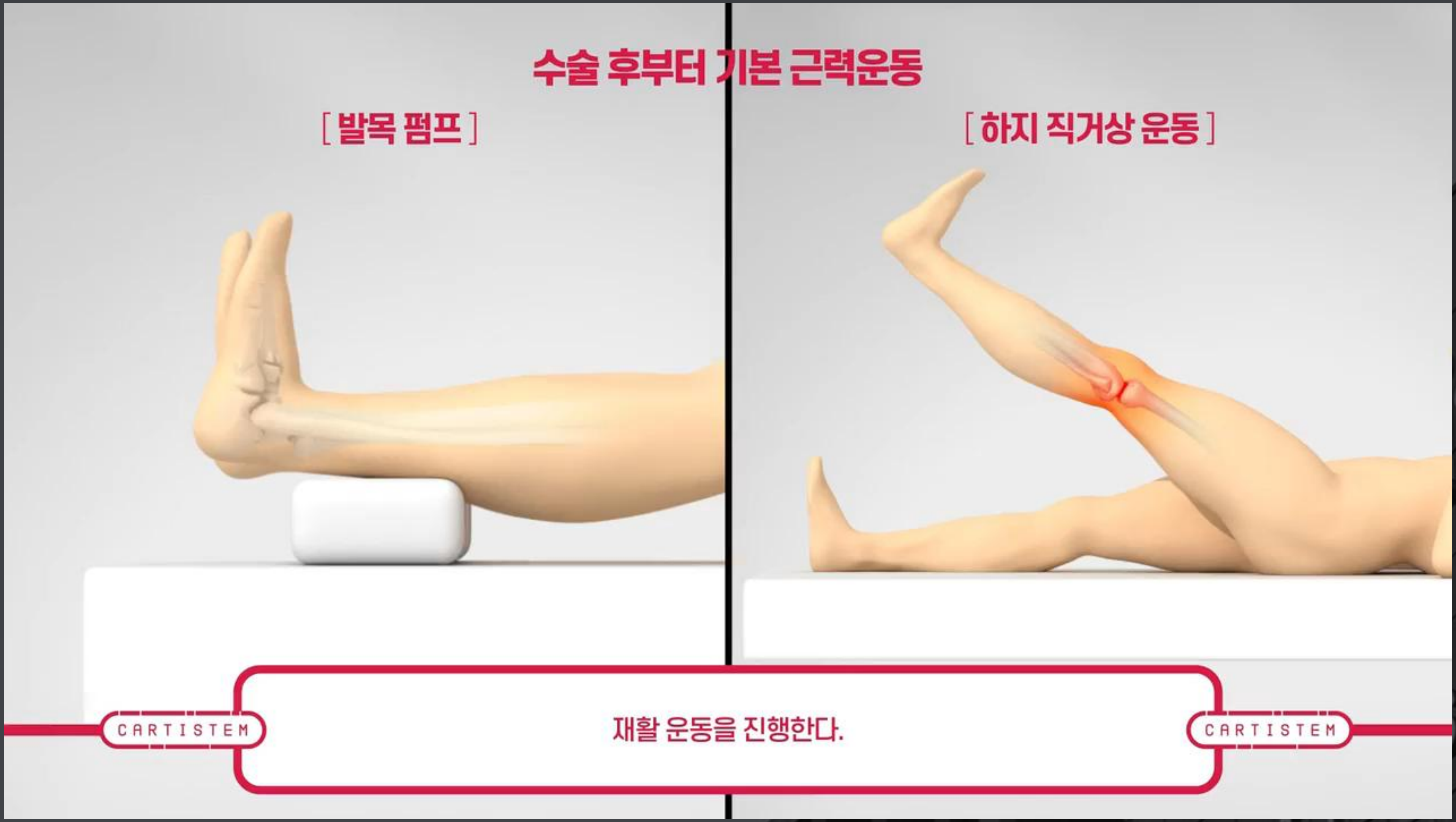
연골 병변 부위에 직경 3~5mm Drill Bit을 이용하여 최대한 촘촘하게
(홀 간의 간격 2~3mm) 각 홀당 5mm 깊이로 홀을 만든다.



카티스템을 연골 병변 부위에 만든 홀에 안쪽(깊은쪽)부터 천천히 채워 넣는다.

카티스템 도포가 끝난 후 투여부위에서 밀려나지 않도록 주의하여
관절에 견인력을 가하며 신전시킨 후 절개 창을 닫고 봉합한다.

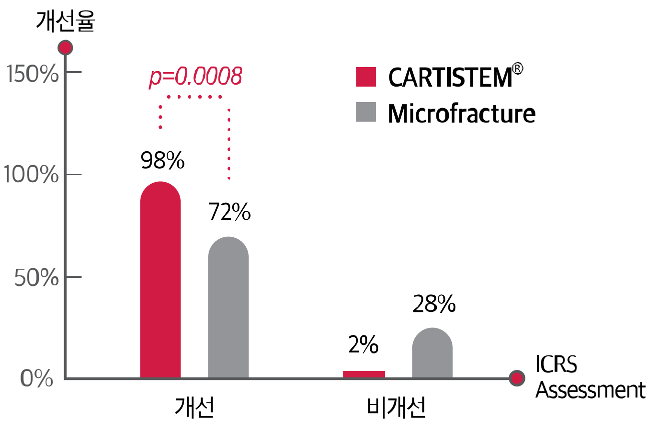
카티스템® 기본 재활법



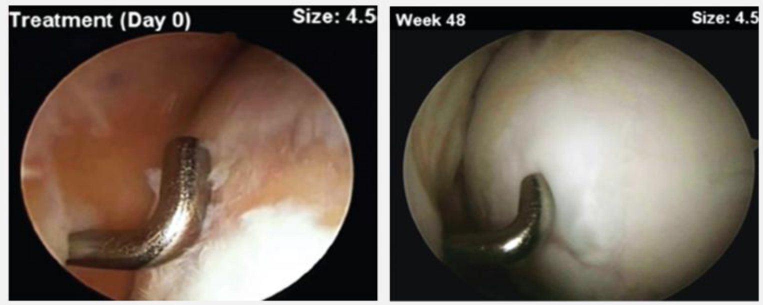
카티스템® 연골재생 효과 확인

ICRS grade Assessment(arthroscopic second look) at 1 years

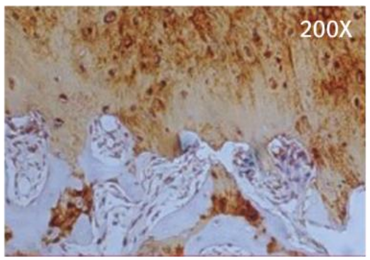
대조시술방법인 Microfracture와 비교한 결과, 일차 유효성 평가 결과인 시술 후 48주째 ICRS overall repair assessment의 개선율은 카티스템® 시술군에서 통계적으로 유의하게 높았습니다(*p-value=0.0008*).



카티스템® : 56세 여자 / 연골결손크기 : 4.5cm²

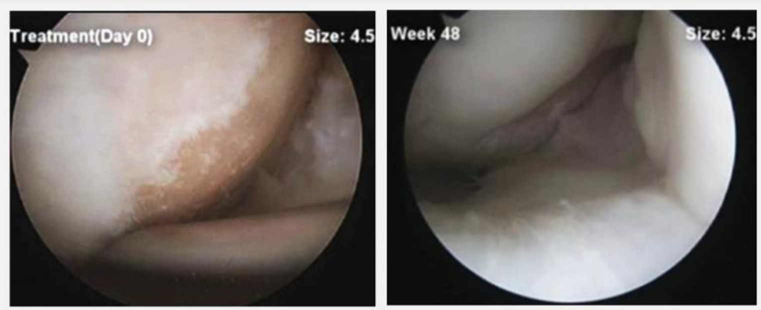


Treatment (Day 0) After 48 weeks

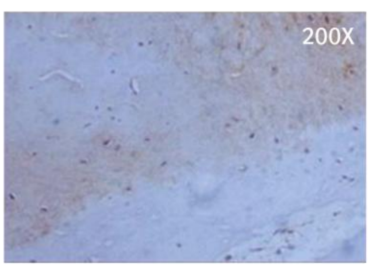


초자연골

미세천공술 : 52세 여자 / 연골결손크기 : 4.5cm²



Treatment (Day 0) After 48 weeks

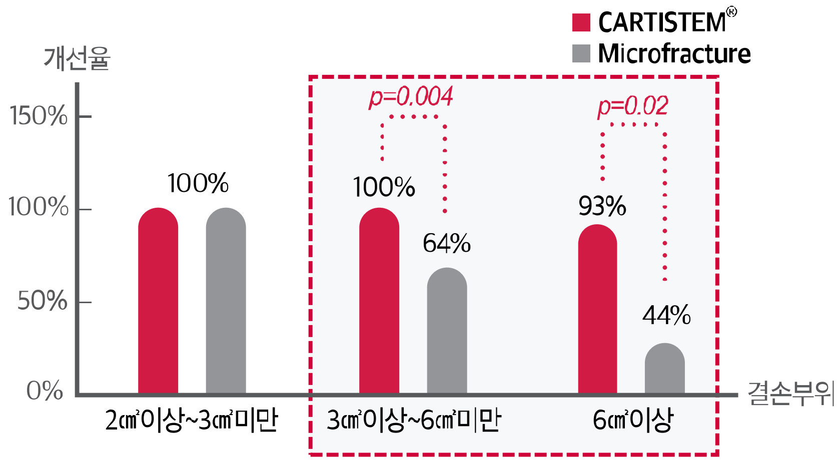


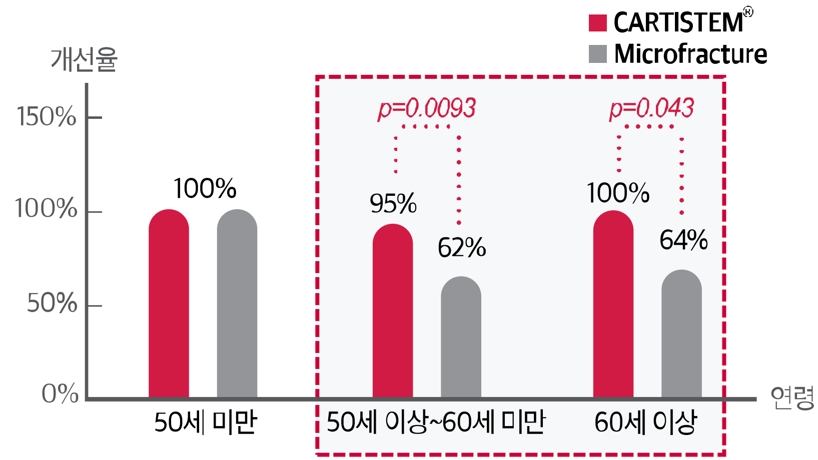
섬유성조직

카티스템® 병변크기 및 연령대별 층화 분석

CARTISTEM

병변크기 및 **연령대별** 층화분석에서 연골재생 효과를 비교한 결과,
카티스템® 시술군에서 대조시술방법인 Microfracture보다 **우수한 연골 재생률**을 확인.

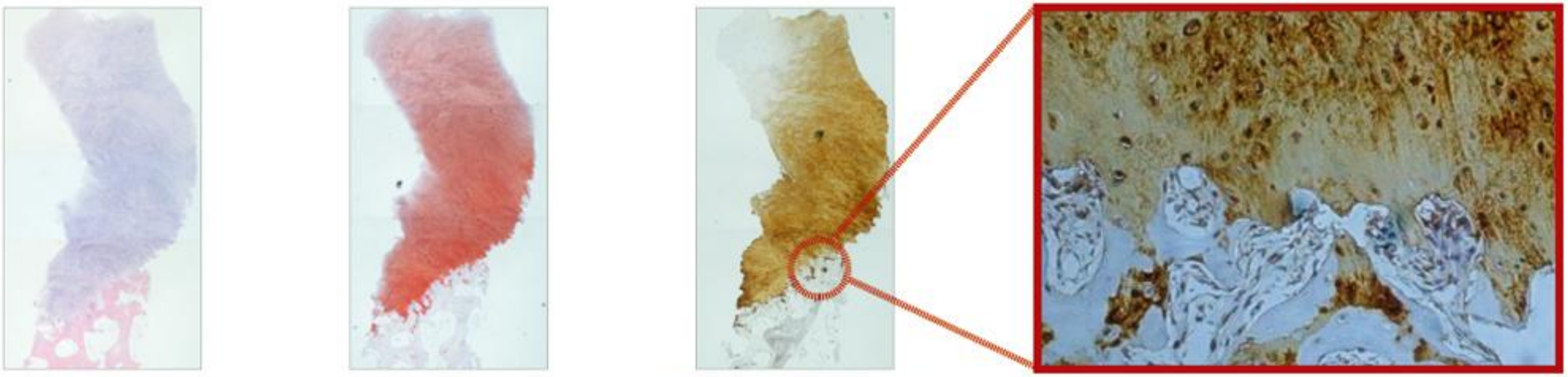




카티스템® 연골재생 효과 확인

카티스템® 조직 생검 염색 결과

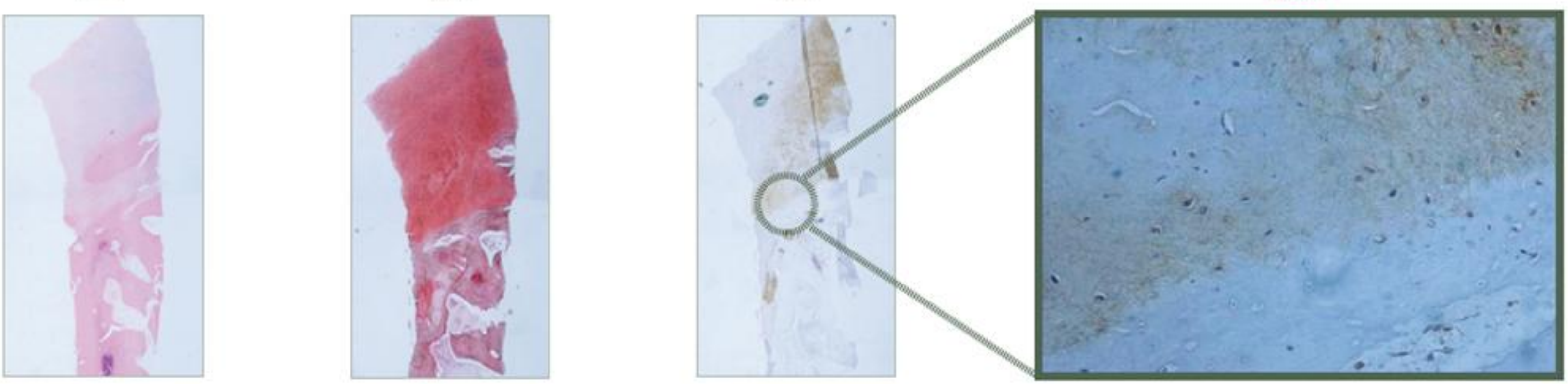
40X | 40X | 40X | 200X



| H&E | | Safranin-O | | Type II Collagen |

미세천공술 조직 생검 염색 결과

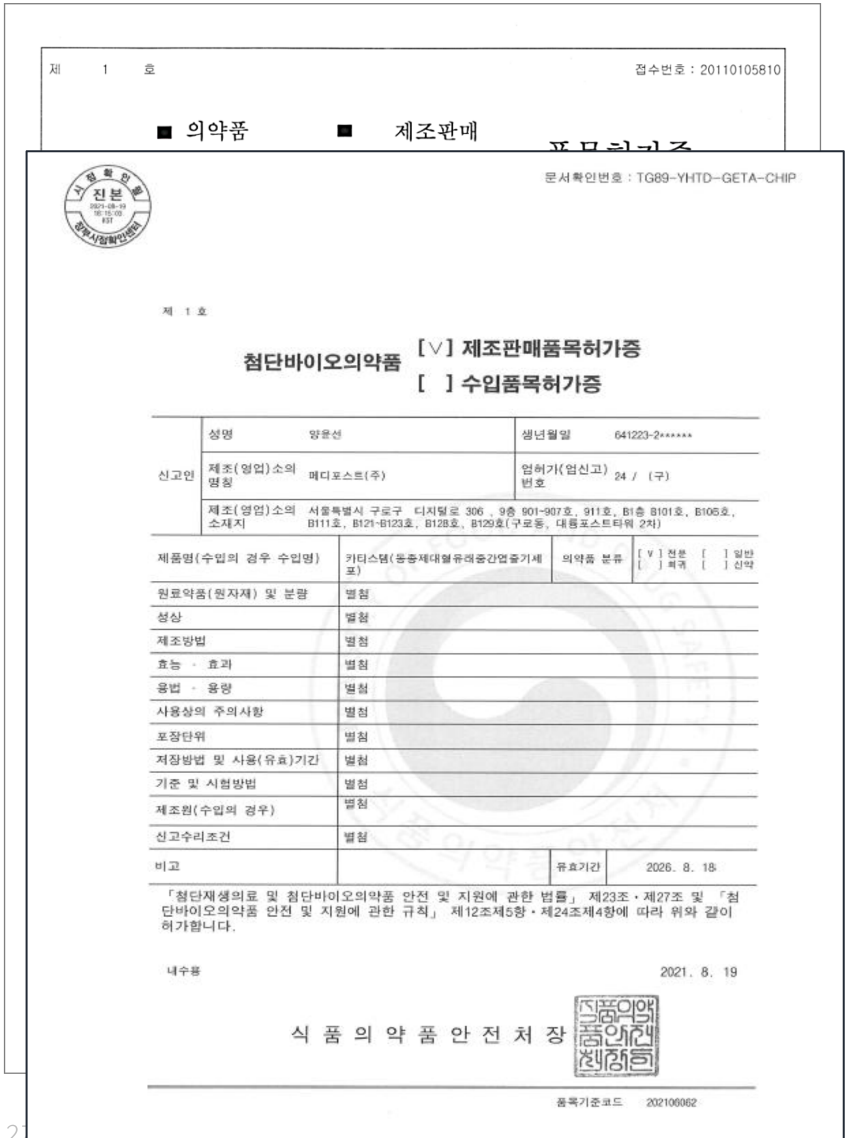
40X | 40X | 40X | 200X



| H&E | | Safranin-O | | Type II Collagen |

카티스템® 식약처 품목허가

CARTISTEM



제 1 호 접수번호 : 20110105810

■ 의약품 ■ 제조판매

문서확인번호 : TG89-YHTD-GETA-CHIP

제 1 호

첨단바이오의약품 [V] 제조판매품목허가증 [ ] 수입품목허가증

신고인	성명	양윤선	생년월일	641223-2******	
	제조(영업)소의 명칭	메디포스트(주)	업허가(업신고) 번호	24 / (구)	
	제조(영업)소의 소재지	서울특별시 구로구 디지털로 306 , 9층 901~907호, 911호, B1층 B101호, B105호, B111호, B121~B123호, B128호, B129호(구로동, 대륭포스트타워 2차)			
제품명(수입의 경우 수입명)		카티스템(동종제대혈유래중간엽줄기세포)	의약품 분류	[V] 전문 [ ] 일반 [ ] 희귀 [ ] 신약	
원료약품(원자재) 및 분량		별첨			
성상		별첨			
제조방법		별첨			
효능 · 효과		별첨			
용법 · 용량		별첨			
사용상의 주의사항		별첨			
포장단위		별첨			
저장방법 및 사용(유효)기간		별첨			
기준 및 시험방법		별첨			
제조원(수입의 경우)		별첨			
신고수리조건		별첨			
비고			유효기간	2026. 8. 18	

「첨단재생의료 및 첨단바이오의약품 안전 및 지원에 관한 법률」 제23조 · 제27조 및 「첨단바이오의약품 안전 및 지원에 관한 규칙」 제12조제5항 · 제24조제4항에 따라 위와 같이 허가합니다.

내수용 2021. 8. 19

식 품 의 약 품 안 전 처 장

품목기준코드 202106062

“퇴행성 또는 반복적 외상으로 인한 골관절염환자 (ICRS Grade IV)의 무릎 연골결손 치료”

❖ Proceedings

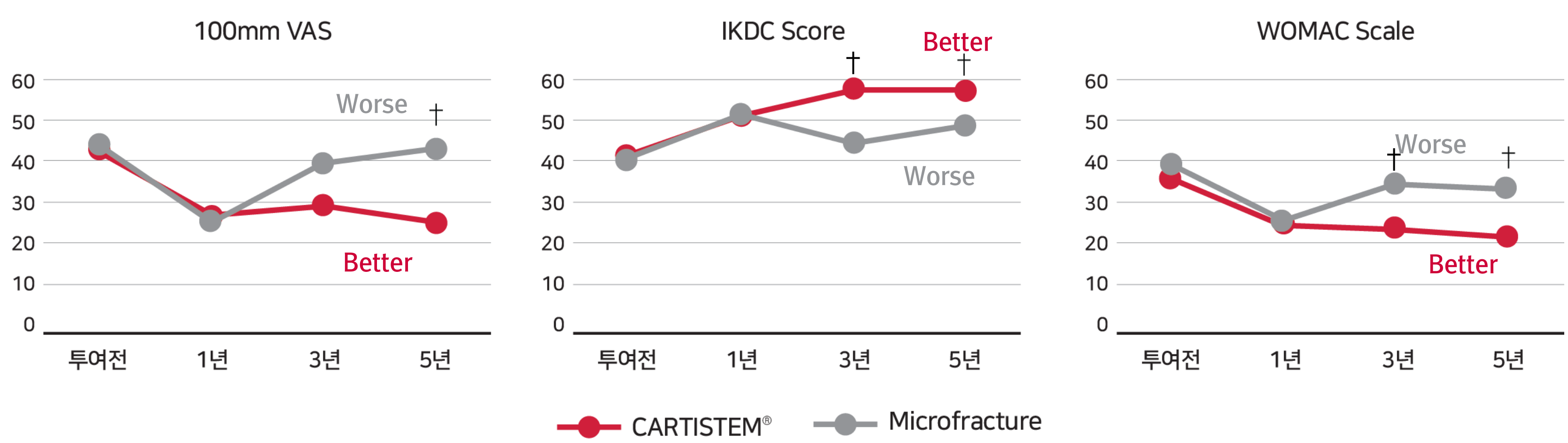
- 임상3상 관찰기간: 2009.02.02~2011.01.24
- 식약처 심사
  ; 안전성 및 유효성
  ; 성분, 제조, 품질관리
  ; GMP 인증
  ; 중앙약심 2011.11.28

➔ **품목허가: 2012년 1월 18일 (전문의약품)**
  –국내 2번째 허가 줄기세포치료제
  –세계 최초 동종제대혈유래줄기세포치료제

- 2012. 5월 시판 시작

카티스템® 임상 3상의 5년차 결과

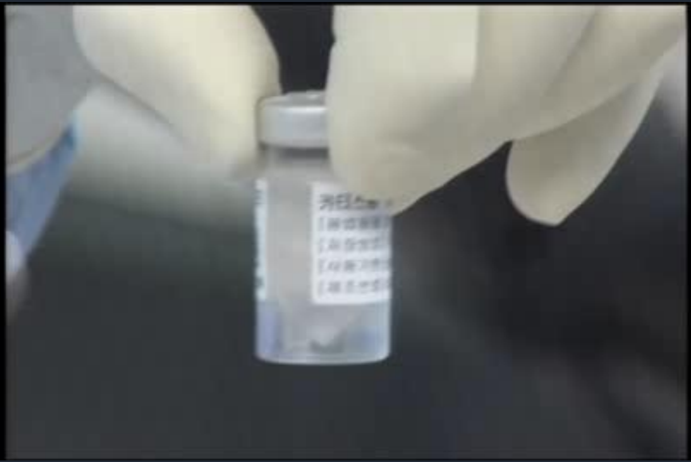
재생된 연골의 기능평가인 100mm VAS, IKDC Score, WOMAC scale 평가를
5년간 추적 관찰한 결과, **카티스템® 시술군**에서 대조시술방법인 Microfracture보다
통계적으로 유의하게 개선 확인(†:p<0.05).



Allogeneic Stem Cell Drug
for Osteoarthritis

- ✓ 1회 수술로 탁월한 연골재생 효과
- ✓ 나이 제한 없이 초자연골 재생 효과
- ✓ 큰 병변에서도 탁월한 연골재생 확인
- ✓ 풍부한 장기 임상 연구자료
- ✓ 골관절염 및 연골결손의 재생치료제

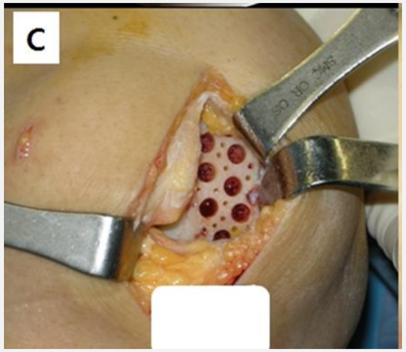
카티스템® 세포 바이알 탭핑

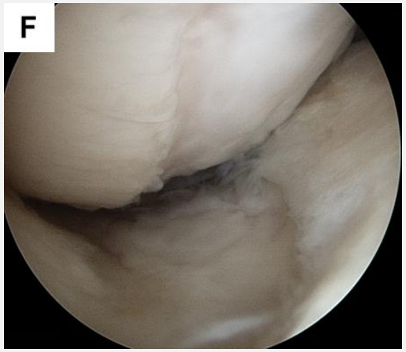


피부절개 및 관절강내 지방 조직 제거

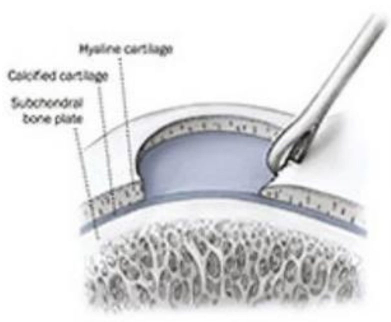
ROM에 방해 되거나 될 수 있는 조직

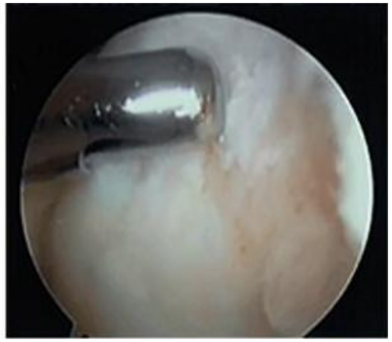
4-5mm 드릴사용 후 K-강선을 이용한 추가 드릴링



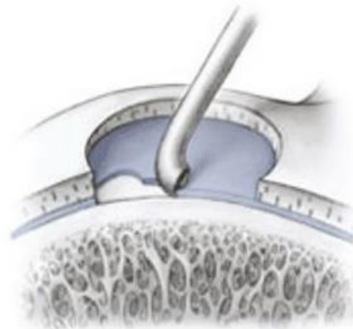


- 석회화(calcified layer) 된 연골 제거 후 드릴 진행 (Subchondral bone의 노출)



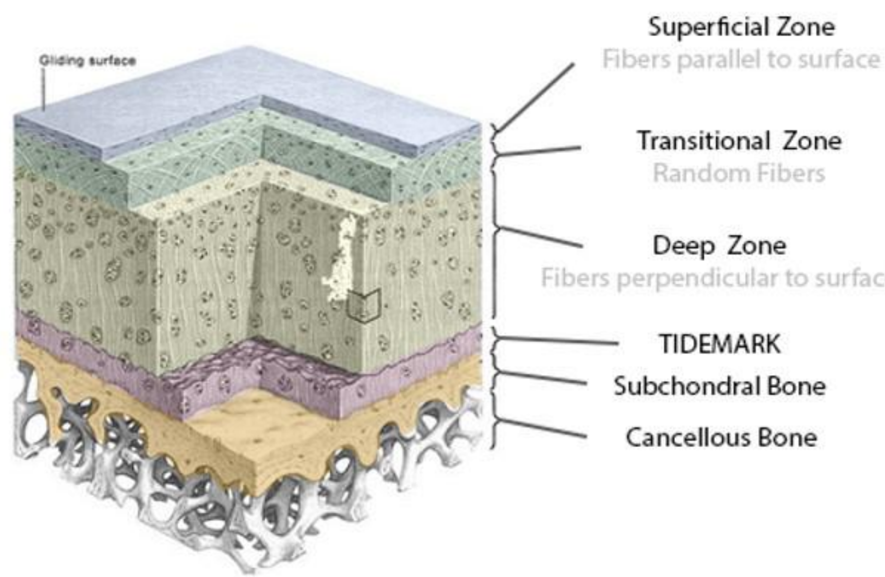


손상된 연골 부위를 정리합니다.



골수강 내 출혈을 유도, 흉터가 치유되듯 뼈와 연골이 함께 재생되도록 합니다.

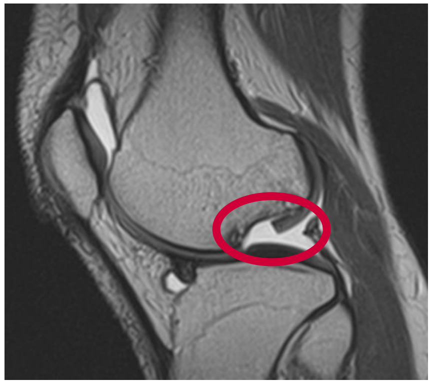
- 손상된 연골의 제거

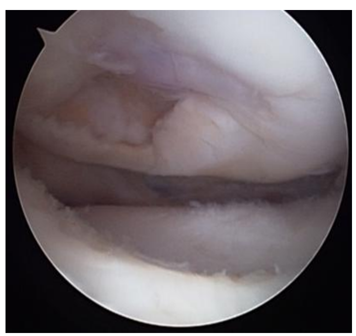


Pre OP

X-ray

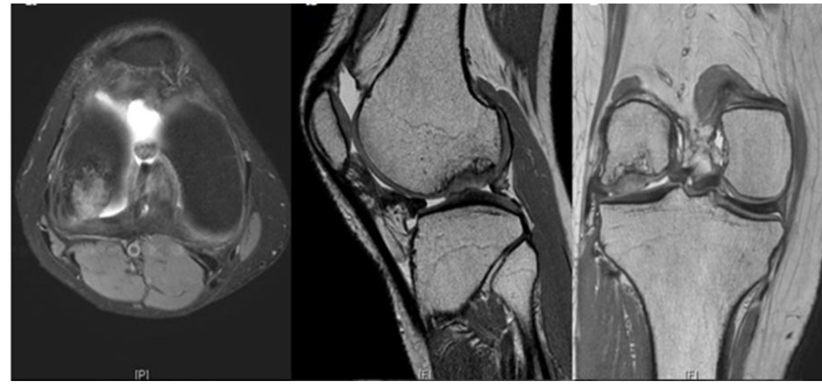
MRI



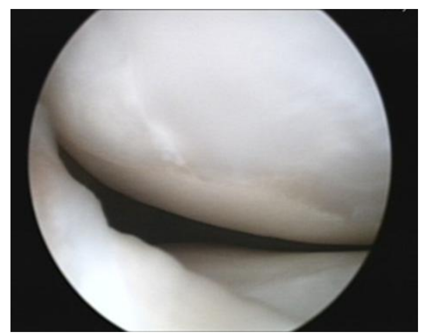


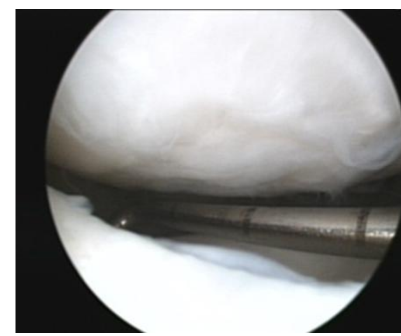
Post OP

1 year MRI



1 year AS

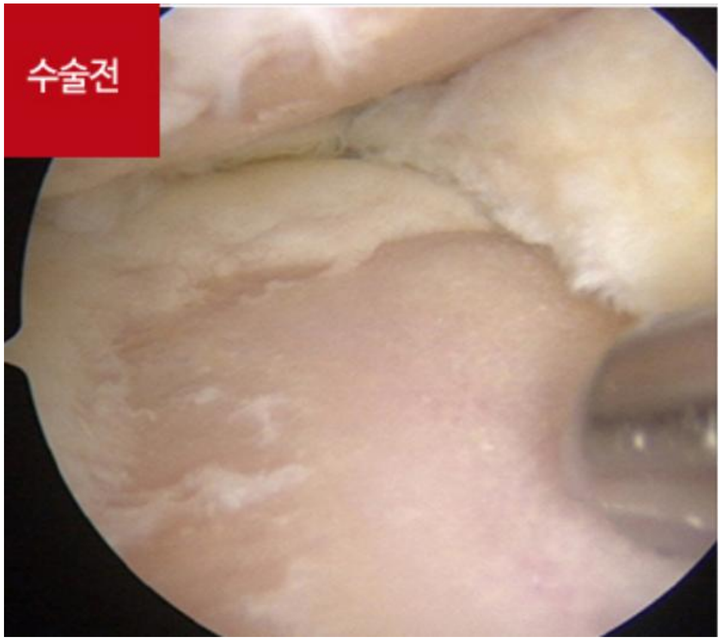




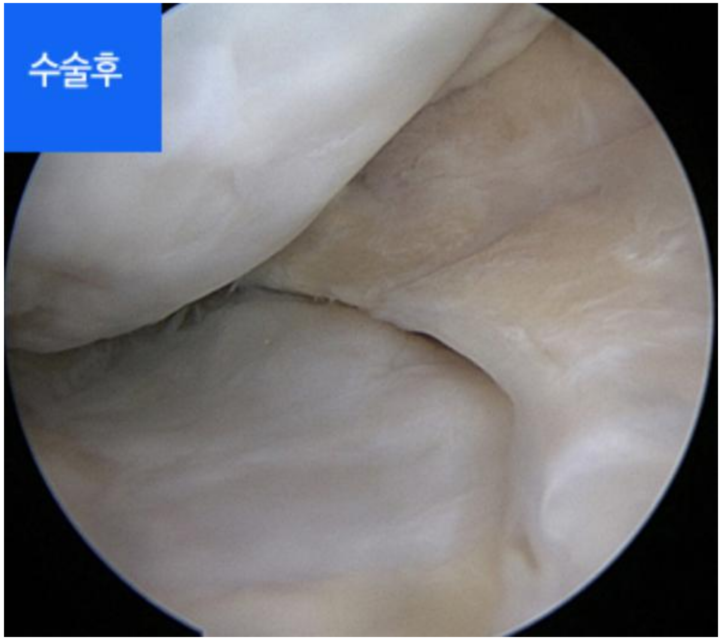
F/32-삼성서울병원, 하철원교수

Pre OP

Post OP



카티스템®
시술 후
13개월

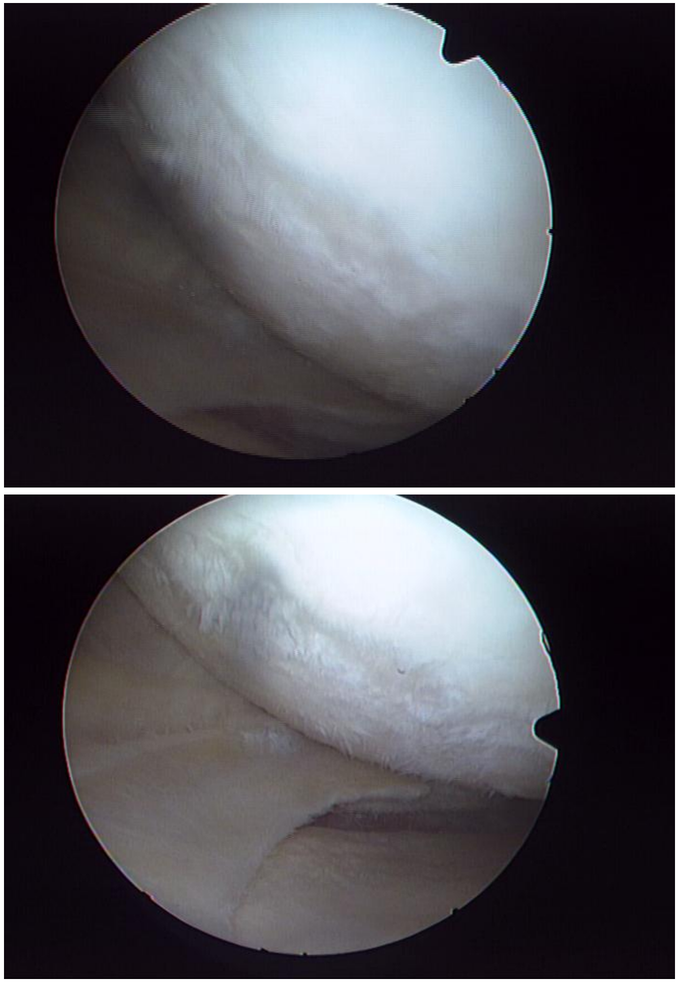


Femur에만 카티스템® 투여 했으나 paracrine effect로
tibia의 결손 부위에도 연골 생성 확인

F/62-서울제이에스병원, 송준섭원장

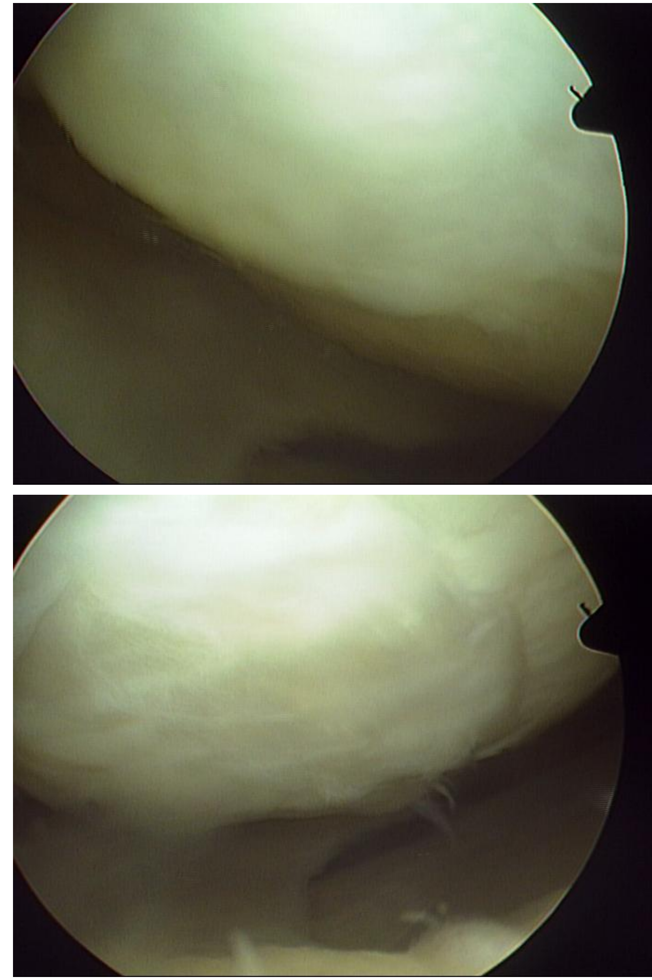
카티스템® Case Report

Pre OP



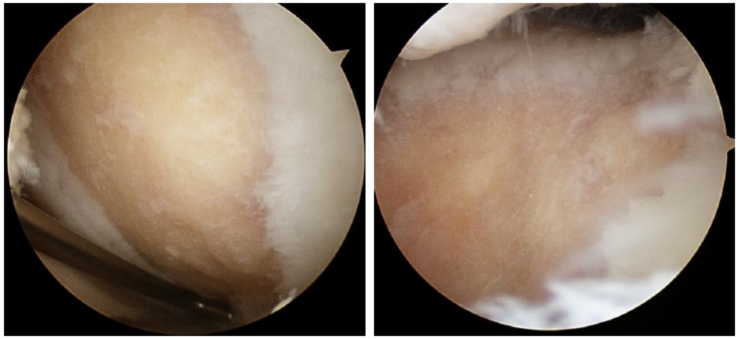
카티스템®
시술 후
12개월

Post OP



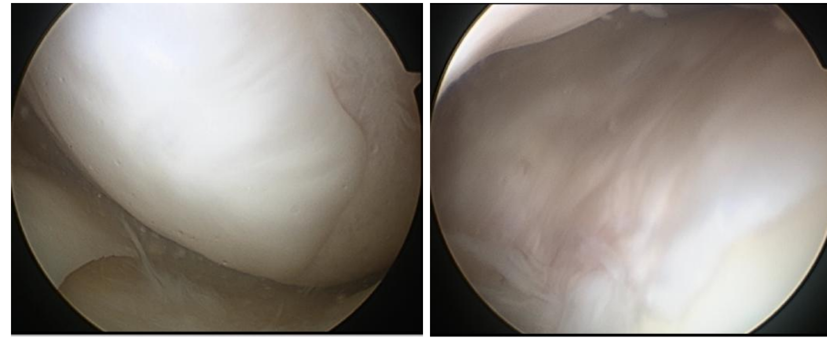
F/55-선정형외과병원, 선승덕 원장

Pre OP

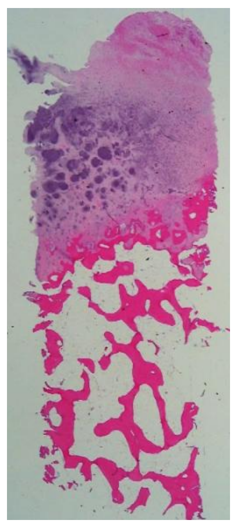


1 year
AS

Post OP

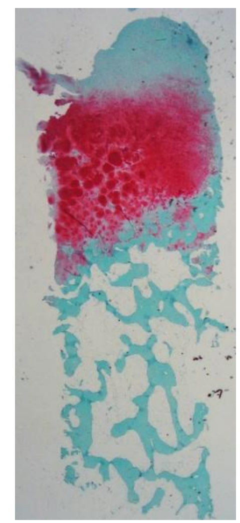


Coll II



H&E

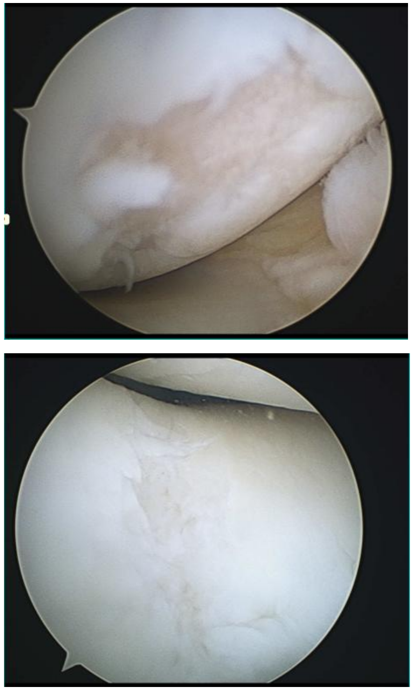
MT



Safranin O

M/56-본서부병원, 권혁남원장

카티스템® Case Report

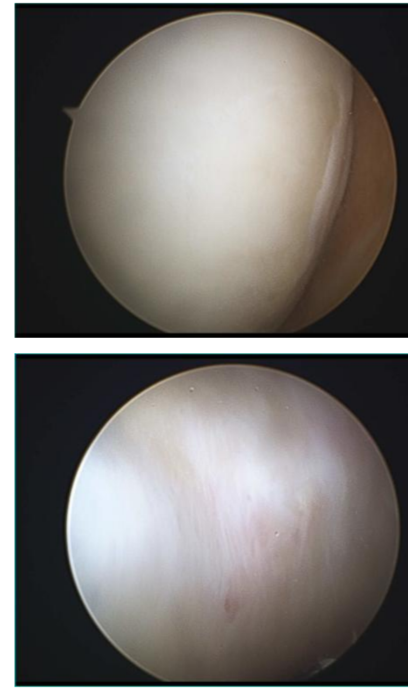


내원 4~5년 전 Rt. knee pain으로 타병원에서 **microfracture** 시행
➔ 증상 유지되고 연골재생 안 되었음. (VAS 6~7).

➔ 재생 실패된 연골 결손부위에 다시 microfracture(MFC, FT)

microfracture(MFC, FT) 6개월 뒤 Arthroscope 2ndary look 시행
연골 재생 실패: 당시 VAS 5

➔ **1개월 뒤 CARTISTEM® 시행**

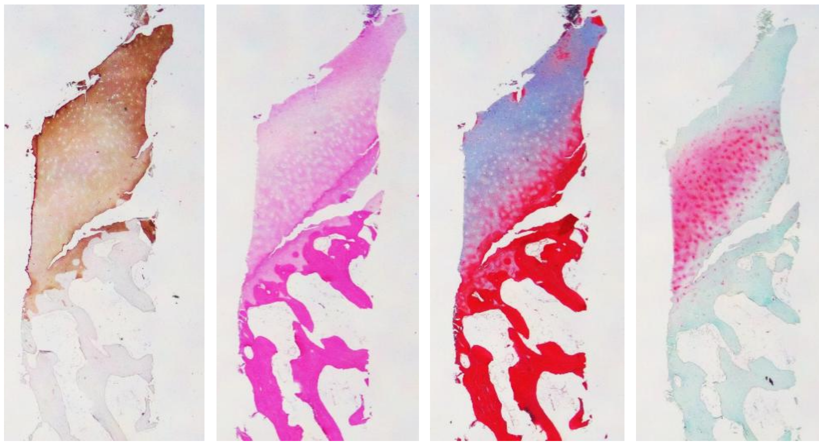


CARTISTEM® 시행 6개월
Arthroscope 2ndary look: 당시 VAS 0 ~ 1

M/47-더드림병원, 도관홍원장

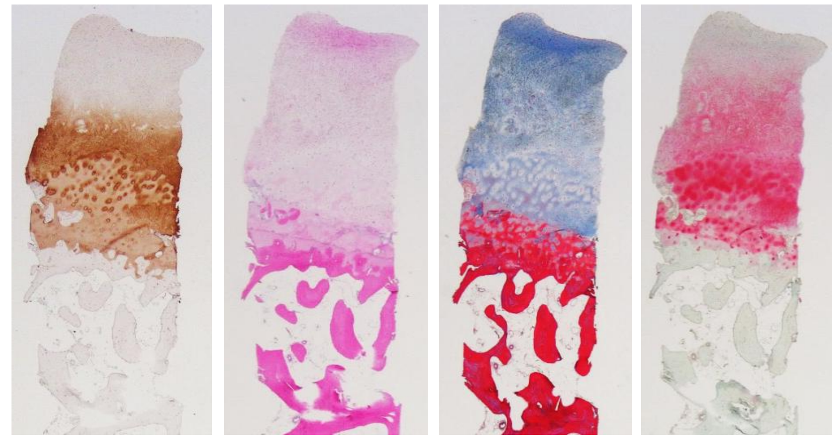
카티스템® Case Report

송** (1649)



Coll II H&E MT Safranin O

최** (0448)



Coll II H&E MT Safranin O

더드림병원, 도관홍원장

CARTISTEM

Published January 12,2021

Original Research

Allogeneic Umbilical Cord Blood–Derived Mesenchymal Stem Cell Implantation Versus Microfracture for Large, Full-Thickness Cartilage Defects in Older Patients

A Multicenter Randomized Clinical Trial and Extended 5-Year Clinical Follow-up

Hong-Chul Lim,[*] MD, Yong-Beom Park,[†] MD, PhD, Chul-Won Ha,[‡§||¶] MD, PhD, Brian J. Cole,[#] MD, MBA, Beom-Koo Lee,[**] MD, Hwa-Jae Jeong,[††] MD, Myung-Ku Kim,[‡‡] MD, Seong-Il Bin,[§§] MD, Chong-Hyuk Choi,[||||] MD, Choong Hyeok Choi,[¶¶] MD, Jae-Doo Yoo,[##] MD, for the Cartistem Research Group[a]

Investigation performed at 10 tertiary-care hospitals in the Republic of Korea

Background: There is currently no optimal method for cartilage restoration in large, full-thickness cartilage defects in older patients.

Purpose: To determine whether implantation of a composite of allogeneic umbilical cord blood–derived mesenchymal stem cells and 4% hyaluronate (UCB-MSC-HA) will result in reliable cartilage restoration in patients with large, full-thickness cartilage defects and whether any clinical improvements can be maintained up to 5 years postoperatively.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: A randomized controlled phase 3 clinical trial was conducted for 48 weeks, and the participants then underwent extended 5-year observational follow-up. Enrolled were patients with large, full-thickness cartilage defects (International Cartilage Repair Society [ICRS] grade 4) in a single compartment of the knee joint, as confirmed by arthroscopy. The defect was treated either with UCB-MSC-HA implantation through mini-arthrotomy or with microfracture. The primary outcome was proportion of participants who improved by $\geq$1 grade on the ICRS Macroscopic Cartilage Repair Assessment (blinded evaluation) at 48-week arthroscopy. Secondary outcomes included histologic assessment; changes in pain visual analog scale (VAS) score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and International Knee Documentation Committee (IKDC) score from baseline; and adverse events.

Results: Among 114 randomized participants (mean age, 55.9 years; 67% female; body mass index, 26.2 kg/m^2), 89 completed the phase 3 clinical trial and 73 were enrolled in the 5-year follow-up study. The mean defect size was 4.9 cm^2 in the UCB-MSC-HA group and 4.0 cm^2 in the microfracture group ($P = .051$). At 48 weeks, improvement by $\geq$1 ICRS grade was seen in 97.7% of the UCB-MSC-HA group versus 71.7% of the microfracture group ($P = .001$); the overall histologic assessment score was also superior in the UCB-MSC-HA group ($P = .036$). Improvement in VAS pain, WOMAC, and IKDC scores were not significantly different between the groups at 48 weeks, however the clinical results were significantly better in the UCB-MSC-HA group at 3- to 5-year follow-up ($P < .05$). There were no differences between the groups in adverse events.

Conclusion: In older patients with symptomatic, large, full-thickness cartilage defects with or without osteoarthritis, UCB-MSC-HA implantation resulted in improved cartilage grade at second-look arthroscopy and provided more improvement in pain and function up to 5 years compared with microfracture.

Registration: NCT01041001, NCT01626677 (ClinicalTrials.gov identifier).

Keywords: full-thickness cartilage defect; cartilage restoration; mesenchymal stem cells; umbilical cord blood; microfracture

The Orthopaedic Journal of Sports Medicine, 9(1), 2325967120973052
DOI: 10.1177/2325967120973052

Articular cartilage defects remain a challenging clinical problem. Currently available treatment options are generally more applicable to localized, focal defects in relatively young

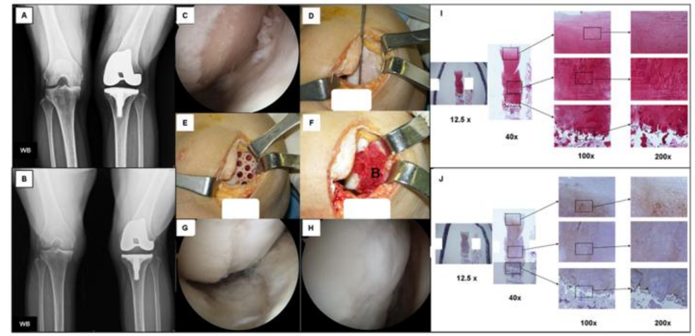


Figure 1. Surgical implantation procedure for the allogeneic human umbilical cord blood–derived mesenchymal stem cell and 4% hyaluronate (UCB-MSC-HA) composite. (A) Preoperative weightbearing (WB) posteroanterior radiograph (Rosenberg view) shows osteophyte formation of the right knee of a 54-year-old female patient. She had previously undergone a total knee arthroplasty for her left knee. (B) The same radiographic view taken at 5-year follow-up shows maintenance of restored medial joint space without significant deterioration. (C) Arthroscopic inspection and confirmation of International Cartilage Repair Society (ICRS) grade 4 cartilage defect at the initial examination. (D) Mini-arthrotomy and measurement of the size of the full-thickness cartilage defect in the medial femoral condyle of the patient's right knee. (E) Multiple drilling with a drill bit of 5-mm diameter, which is larger than the diameter of holes made during standard microfracture at the subchondral bone. The large drill holes were mainly made for containment of the UCB-MSC-HA composite. (F) Implantation of the UCB-MSC-HA composite in the 5-mm drill holes. (G, H) Restored articular cartilage at 48 weeks after implantation, which was assessed as ICRS grade 3 in this case. (I) Safranin O staining showing abundant presence of glycosaminoglycan in the restored cartilaginous tissue. (J) Type II collagen immunostaining showing abundant type II collagen in the restored cartilaginous tissue.

Outcome Measures

The primary outcome of the phase 3 clinical trial was the proportion of participants with cartilage restoration equivalent to at least 1 grade improvement on the ICRS Macroscopic Cartilage Repair Assessment at 48-week arthroscopic evaluation (see Appendix Table A2 for scoring).[43] Secondary outcomes were the ICRS II Histological Evaluation System score from tissue biopsies[23,24] and changes in 100-mm visual analog scale (VAS) for pain, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC),[2] and International Knee Documentation Committee (IKDC) scores[15] from baseline to 48 weeks. Biopsies of the repair tissue taken from the center of the lesion site during the 48-week arthroscopic evaluation were used for the histologic assessment.

Owing to differences in the surgical scars, the ICRS Macroscopic Cartilage Repair Assessment was conducted in a blinded fashion as follows: Arthroscopic images were captured and videos were recorded during both the initial surgical procedure and the second-look arthroscopy for efficacy evaluation at 48 weeks. Ten investigators were divided into 2 groups (group 1: HC Lim, BK Lee, HJ Jeong, CH Choi, CW Ha; group 2: MK Kim, SI Bin, CH Choi, JD Yoo, JR Yoon), and each group assessed the arthroscopic images and videos of the other group without any information regarding the treatment assignment. The 5 investigators in each group initially assessed preoperative and postoperative images and videos independently using the ICRS Macroscopic Cartilage Repair Assessment. The evaluations were collected and compared to arrive at a final consensus. If agreement was reached by at least 3 of the 5 reviewers, the 3 matching results were selected as the final ICRS grade. For cases of agreement by <3 reviewers, the final grade was determined through an open discussion among the 5 investigators in each group.

The WOMAC is a well-validated, disease-specific measure of osteoarthritis of the hip or knee that includes pain, stiffness, and function subscales and addresses activities of daily living.[2] The IKDC subjective knee form[15] is designed to measure symptoms and limitations in function and sports activity for various knee conditions, including cartilage injuries, and has demonstrated strong psychometric characteristics. The IKDC has shown adequate internal consistency and no remarkable floor or ceiling effects.[5]

Orthop J Sports Med 2021 Jan 12;9(1):2325967120973052

KNEE

Knee Surgery, Sports Traumatology, Arthroscopy WILEY

Implantation of hUCB-MSCs generates greater hyaline-type cartilage than microdrilling combined with high tibial osteotomy

Se-Han Jung[1,2] | Hyunjin Park[3] | Min Jung[1,4] | Kwangho Chung[1,5] | Sungjun Kim[6] | Hyun-Soo Moon[1,4] | Jisoo Park[1,2] | Ju-Hyung Lee[2] | Chong-Hyuk Choi[1,4] | Sung-Hwan Kim[1,2]

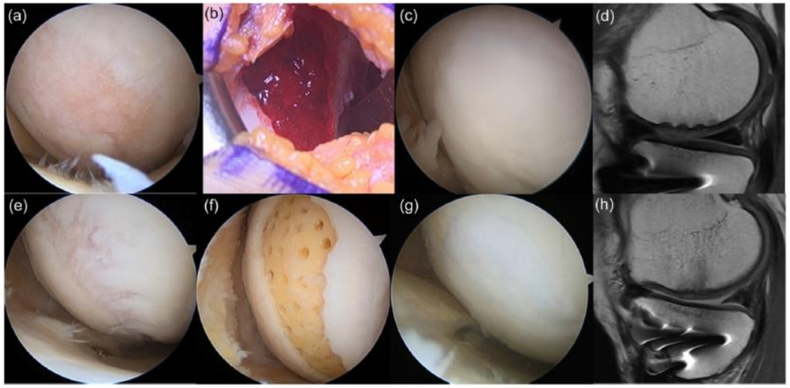


FIGURE 1 Implantation of human allogeneic umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) and arthroscopic microdrilling. (a–d) Implantation of hUCB-MSCs. (e–h) Arthroscopic microdrilling. (a, e) Preoperative cartilage defect on medial femoral condyle viewed from anterolateral portal. (b) After hUCB-MSC implantation through a mini-arthrotomy. (c) Repaired cartilage after hUCB-MSC implantation on second-look arthroscopy. (d) Favourable cartilage regeneration on magnetic resonance imaging, except for subchondral bone oedema and minimal bony defect. (f) After arthroscopic microdrilling. (g) Repaired cartilage after microdrilling on second-look arthroscopy. (h) Favourable cartilage regeneration on magnetic resonance image, except for subchondral bone oedema.

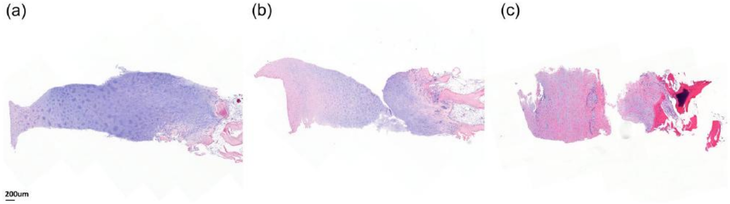


FIGURE 3 Subgrouping into three groups depending on hyaline or fibrocartilage formation. (a) Predominantly hyaline cartilage (H group) shows homogeneous basophilic material deposition in the mid/deep zone. (b) Fibrocartilage-hyaline mixture (F–H group) shows a heterogeneous mixture of basophilic and eosinophilic material deposition. (c) Fibrocartilage (F group) predominantly shows eosinophilic collagen deposition.

TABLE 3 Cartilage regeneration assessment on second-look arthroscopy.[a]

Evaluation metrics	Study group ($n = 15$)	Control group ($n = 10$)	p Value
ICRS CRA overall points	8.5 ± 2.1	7.7 ± 3.4	0.800
Defect repair	3.7 ± 0.5	3.0 ± 1.4	0.263
Integration	2.1 ± 1.1	1.8 ± 1.0	0.578
Macroscopic appearance	2.8 ± 0.9	2.9 ± 1.3	0.707
ICRS CRA grade			0.327
Grade I (normal)	2 (13.3)	1 (10)	1.000
Grade II (near normal)	8 (53.3)	5 (50)	1.000
Grade III (abnormal)	5 (33.3)	2 (20)	0.659
Grade IV (severely abnormal)	0 (0)	2 (20)	0.150
Total coverage, %	68.0 ± 20.6	62.0 ± 27.8	0.445
Total covered area, cm^2	4.7 ± 1.5	3.0 ± 1.6	0.020*
Rigidity of repaired tissue			0.132
Rigid	13 (86.7)	5 (50)	0.075*
Softer	1 (6.7)	3 (30)	0.267
Very soft	1 (6.7)	2 (20)	0.543

Abbreviations: hUCB-MSCs, human umbilical cord blood-derived mesenchymal stem cells; ICRS CRA, International Cartilage Repair Society cartilage repair assessment.
[a]Values are presented as mean ± SD or number of patients (%).
*Statistically significant ($p < 0.05$) or borderline significance ($0.05 \le p < 0.1$).

TABLE 5 Histological assessment.[a]

Overall tissue assessment			0.041*
Predominantly hyaline (H)	10 (66.7)	2 (20)	
Fibrocartilage-hyaline mixture (F-H)	4 (26.7)	4 (40)	
Fibrocartilage (F)	1 (6.7)	4 (40)	

Abbreviation: ICRS, International Cartilage Repair Society.
[a]Values are presented as mean ± SD or number of patients (%).
*Statistical significance ($p < 0.05$).

Bae *et al. Stem Cell Research & Therapy* (2025) 16:211
https://doi.org/10.1186/s13287-025-04356-9
Stem Cell Research & Therapy

RESEARCH Open Access

Treatment of osteoarthritic knee with high tibial osteotomy and allogeneic human umbilical cord blood–derived mesenchymal stem cells combined with hyaluronate hydrogel composite

Bo Seung Bae[1], Jae Woong Jung[1], Gyeong Ok Jo[1], Seon Ae Kim[1], Eun Jeong Go[1], Mi-La Cho[2], Asode Ananthram Shetty[3] and Seok Jung Kim[1,4*]

Table 2 Comparison of preoperative and 2-year postoperative statuses based on arthroscopic and radiologic evaluations

	Preoperative ($n = 10$)	Postoperative ($n = 10$)	Difference	p
Lesion size (cm^2)	7.00 [4.38–10.50]	0.16 [0.00–1.75]	6.00 [4.38–7.81]	0.002[a]
HKA angle (°)	7.50 [7.00–10.25]	-1.00 [-3.5–0.00]	10.50 [8.5–11.25]	0.002[a]
ICRS grade (n)	4 [4–4]	1 [1–2.25]	3 [1.75–3]	0.002[a]
KL grade (n)	3 [3–3]	2 [2–2]	1 [1–1]	0.004[a]

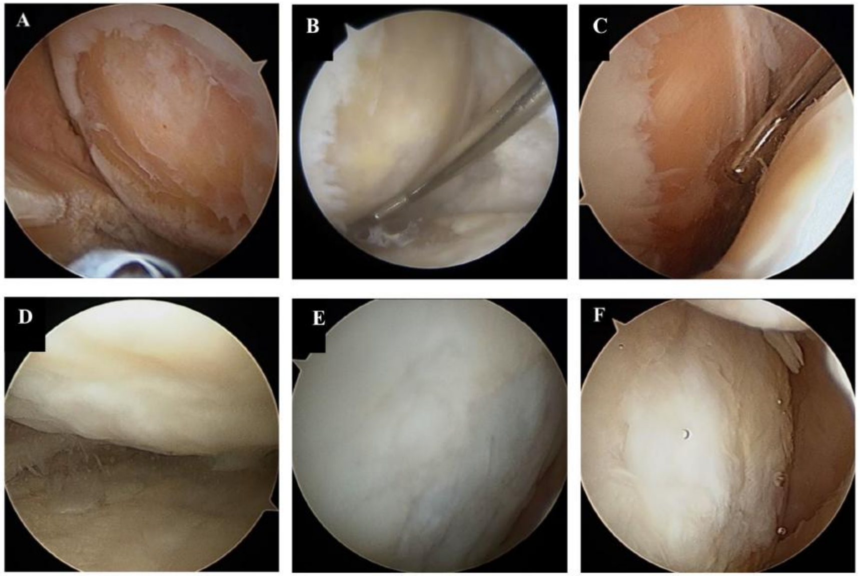


Fig. 3 Comparison between preoperative and second-look arthroscopy. (**A**, **D**) Chondral lesion of MFC in a 60-year-old male patient preoperatively (**A**) and 2-year postoperatively (**D**). (**B**, **E**) Chondral lesion of MFC in a 59-year-old male patient preoperatively (**B**) and 2-year postoperatively (**E**). (**C**, **F**) Chondral lesion of MFC in a 57-year-old female patient preoperatively (**C**) and 2-year postoperatively (**F**). MFC, medial femoral condyle

Table 3 Comparison between preoperative and postoperative clinical outcomes

-	WOMAC	p	VAS	p
Preoperative	57.00 [44.75–63.00]	< 0.001[a]	66.25 [42.25–75.50]	< 0.001[a]
6-month postoperative	47.50 [35.50–54.50]	0.189[b]	38.75 [25.50–58.50]	0.002[b]
1-year postoperative	36.0 [28.50–47.75]	0.004[c]	38.00 [16.63–48.50]	0.002[c]
2-years postoperative	27.50 [21.50–29.50]	0.002[d]	26.25 [13.63–32.38]	0.002[d]
-	SF-36 PCS	p	SF-36 MCS	p
Preoperative	27.97 [25.39–37.34]	< 0.001[a]	41.04 [26.33–53.98]	0.014[a]
6-month postoperative	43.75 [29.84–48.98]	0.037[b]	55.94 [38.13–59.43]	0.064[b]
1-year postoperative	48.91 [35.55–59.29]	0.002[c]	54.06 [40.65–66.98]	0.049[c]
2-years postoperative	55.31 [49.38–63.75]	0.002[d]	63.18 [53.67–66.33]	0.020[d]

logy, MEDIPOST

Contents lists available at ScienceDirect

Orthopaedics & Traumatology: Surgery & Research

journal homepage: www.sciencedirect.com/journal/orthopaedics-and-traumatology-surgery-and-research

Original article

Effect of preoperative medial meniscus status on the outcomes of high tibial osteotomy with human umbilical cord-derived mesenchymal stem cells cartilage regeneration

Dhong-Won Lee [a,*], Sung-Wook Hong [a], Seung-Ik Cho [b], Sung-Gyu Moon [c], Ji-Hee Kang [c]

[a] Department of Orthopaedic Surgery, KonKuk University Medical Center, Konkuk University School of Medicine, Seoul, Korea
[b] Sports Medical Center, KonKuk University Medical Center, Konkuk University School of Medicine, Seoul, Korea
[c] Department of Radiology, KonKuk University Medical Center, Konkuk University School of Medicine, Seoul, Korea

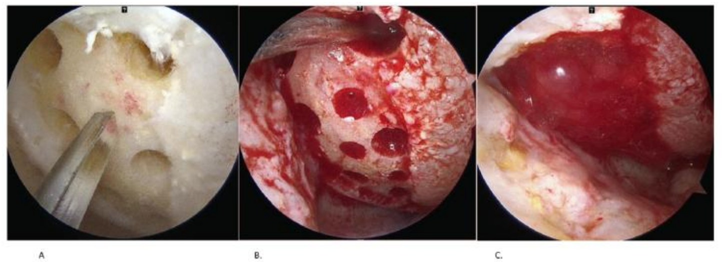


Fig. 2. The process of arthroscopic implantation of human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs). **A:** Multiple drill holes were created at the cartilage defect sites of the medial femoral using a customized drill bit (Medipost Inc., Seoul, Republic of Korea). The drill holes (diameter, 2-3 mm; depth, 10 mm) were spaced approximately 2 mm apart. **B:** The hUCB-MSCs and hyaluronic acid hydrogel composite was implanted into the drill holes from the base to the surface. **C:** The transplanted hUCB-MSCs were confirmed to remain in place even after range of motion.

Table 1
Demographic and baseline characteristics of patients in groups P and L.

	Group P (n = 25)	Group L (n = 22)	p-value
Age, years	56.8 ± 5.4	57.6 ± 5.9	0.553
BMI, kg/m^2	25.1 ± 2.2	26.2 ± 2.9	0.147
Sex, n (male/female)	6/19	7/15	0.550
Clinical follow-up period, months	30.5 ± 4.7	31.3 ± 3.9	0.532
MRI follow-up duration, months	13.1 ± 1.8	12.7 ± 1.6	0.428
Second-look and plate removal, months	13.1 ± 1.6	12.6 ± 1.7	0.305
Preoperative IKDC subjective score	52.3 ± 11.3	48.7 ± 12.4	0.303
Preoperative WOMAC score	45.6 ± 6.4	47.2 ± 6.1	0.387
Preoperative mechanical axis (hip-knee-ankle angle), °	Varus 6.6 ± 1.1	Varus 7.1 ± 1.2	0.154
Preoperative Kellgren–Lawrence grade (I/II/III/IV), n	0/14/11/0	0/9/13/0	0.302
Preoperative ICRS grade of MFC, n (%)			0.901
Grade 3C	5 (20%)	4 (18.2%)	
Grade 4A	18 (72%)	15 (70.2%)	
Grade 4B	2 (8%)	3 (10.6%)	
MFC lesion size, cm^2	6.2 ± 2.3	6.9 ± 2.1	0.2854
Preoperative MM extrusion, mm	3.4 ± 0.6	4.3 ± 1.2	0.002
Preoperative pathological MM extrusion, n (%)	16 (64%)	19 (86.4%)	0.079

Table 2
Clinical and radiological outcomes of patients in groups P and L.

	Group P (n = 25)	Group L (n = 22)	p-value
IKDC subjective score	77.3 ± 10.7	73.1 ± 11.8	0.207
WOMAC score	15.4 ± 7.3	17.2 ± 7.7	0.415
Isokinetic extensor strength at 60 ° (Nm/kg)	121.5 ± 24.3	118.2 ± 28.4	0.669
LSI for extensor strength (%)	0.8 ± 0.2	0.8 ± 0.3	0.999
Mechanical axis (hip-knee-ankle angle), °	Valgus 1.8 ± 1.0	Valgus 2.1 ± 1.1	0.393
Postoperative MM extrusion, mm	4.2 ± 1.1	5.3 ± 1.3	0.016
Pathologic MM extrusion, n (%)	19 (76%)	22 (100%)	0.023

IKDC, International Knee Documentation Committee; WOMAC, Western Ontario and McMaster Universities Osteoarthritis Index; LSI, limb symmetry index; MM, medial meniscus.

Table 3
Cartilage regeneration of the medial femoral condyle on mri and second-look arthroscopy.

	Group P (n = 25)	Group L (n = 22)	p-value
MOCART 2.0 score	56.8 ± 10.5	54.1 ± 8.8	0.347
ICRS CRA score	9.4 ± 1.8	8.6 ± 1.7	0.083
ICRS CRA grade, n (%)			0.626
Grade I	4 (16%)	2 (9.1%)	
Grade II	17 (68%)	14 (63.6%)	
Grade III	4 (16%)	6 (27.3%)	
Grade IV	0	0	
ICRS CRA grade I or II, n (%)	21 (84%)	16 (72.7%)	0.480
Rigidity of repaired tissue, n (%)			0.565
Rigid	17 (68%)	12 (54.5%)	
Softer	7 (28%)	8 (36.4%)	
Very soft	1 (4%)	2 (9.1%)	

MRI, magnetic resonance imaging; MOCART, Magnetic Resonance Observation of Cartilage Repair Tissue; ICRS CRA, International Cartilage Repair Society Cartilage Repair Assessment.

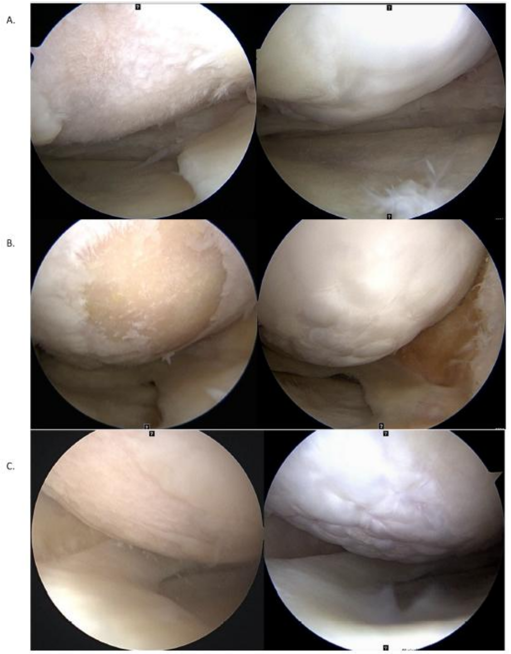


Fig. 3. Evaluation of repaired cartilage according to International Cartilage Repair Society (ICRS) cartilage repair assessment (CRA). A: A case of ICRS CRA score of 12, which was grade I (normal) in group L. B: A case of ICRS CRA score of 10, which was grade II (nearly normal) in group P. C: A case of ICRS CRA score of 6, which

5. Conclusion

HTO with hUCB-MSC implantation provided significant clinical improvements and effective cartilage regeneration regardless of preoperative MM status. However, preoperative MM extrusion may influence subchondral bone changes, emphasizing the need to consider MM status for long-term outcomes.

Contents lists available at ScienceDirect

Regenerative Therapy

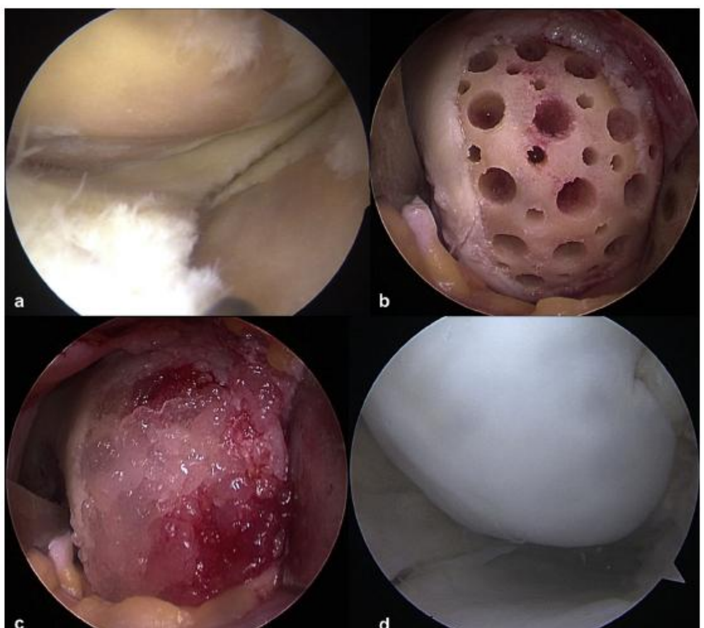
journal homepage: http://www.elsevier.com/locate/reth

Original Article

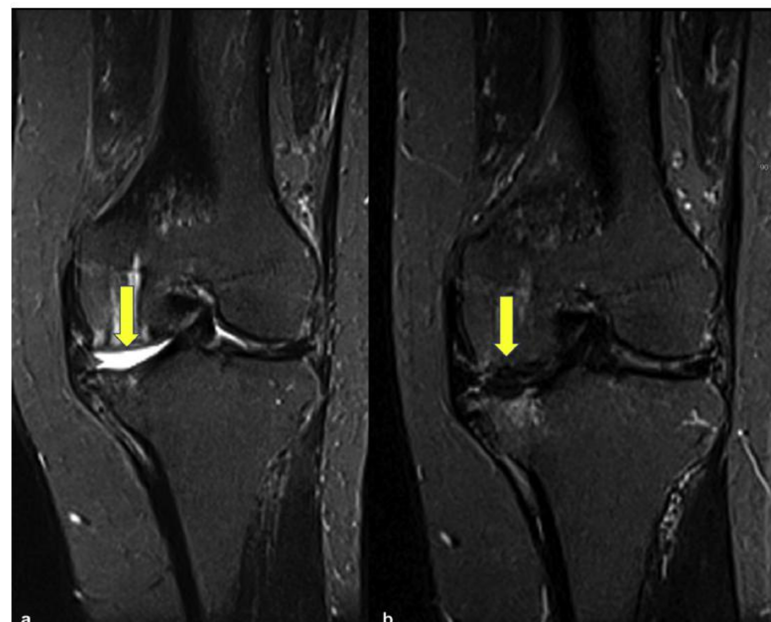
Implantation of allogenic umbilical cord blood-derived mesenchymal stem cells improves knee osteoarthritis outcomes: Two-year follow-up

Jun-Seob Song [a], Ki-Taek Hong [a], Na-Min Kim [a], Jae-Yub Jung [a], Han-Soo Park [a], Sang Heon Lee [b], Yoon Joo Cho [b], Seok Jung Kim [b,*]

[a] Department of Orthopaedic Surgery, Gangnam JS Hospital, Seoul, Republic of Korea
[b] Department of Orthopaedic Surgery, College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea



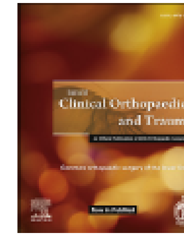
Pre op   Post op



Contents lists available at ScienceDirect

Journal of Clinical Orthopaedics and Trauma

journal homepage: www.elsevier.com/locate/jcot



Allogenic umbilical cord blood-derived mesenchymal stem cells implantation for the treatment of juvenile osteochondritis dissecans of the knee

Jun-Seob Song [a], Ki-Taek Hong [a], Na-Min Kim [a], Jae-Yub Jung [a], Han-Soo Park [a], Yoo-Chang Kim [b], Asode Ananthram Shetty [c], Seok Jung Kim [b,*]

[a] Department of Orthopaedic Surgery, Seoul JS Hospital, Seoul, Republic of Korea
[b] Department of Orthopaedic Surgery, College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea
[c] Canterbury Christ Church University, Faculty of Health and Social Sciences, 30 Pembroke Court, Chatham Maritime, Kent, ME4 4UF, United Kingdom

Pre op

Post op

Papers Published

STEM CELLS
REGENERATIVE MEDICINE

Thrombospondin-2 Secreted by Human Umbilical Cord Blood-Derived Mesenchymal Stem Cells Promotes Chondrogenic Differentiation

PLOS | ONE

RESEARCH ARTICLE

Effect of Transplanting Various Concentrations of a Composite of Human Umbilical Cord Blood-Derived Mesenchymal Stem Cells and Hyaluronic Acid Hydrogel on Articular Cartilage Repair in a Rabbit Model

Yong-Beom Park[1], Chul-Won Ha[2,3,4]*, Jin-A Kim[2,3], Ji-Heon Rhim[2,3], Yong-Geun Park[5]

Original Research

Allogeneic Umbilical Cord Blood–Derived Mesenchymal Stem Cell Implantation Versus Microfracture for Large, Full-Thickness Cartilage Defects in Older Patients

A Multicenter Randomized Clinical Trial and Extended 5-Year Clinical Follow-up

Hong-Chul Lim,* MD, Yong-Beom Park,† MD, PhD, Chul-Won Ha,‡§||¶ MD, PhD, Brian J. Cole,# MD, MBA, Beom-Koo Lee,** MD, Hwa-Jae Jeong,†† MD, Myung-Ku Kim,‡‡ MD, Seong-Il Bin,§§ MD, Chong-Hyuk Choi,|||| MD, Choong Hyeok Choi,¶¶ MD, Jae-Doo Yoo,## MD, for the Cartistem Research Group*

Investigation performed at 10 tertiary-care hospitals in the Republic of Korea

Background: There is currently no optimal method for cartilage restoration in large, full-thickness cartilage defects in older patients.

Purpose: To determine whether implantation of a composite of allogeneic umbilical cord blood–derived mesenchymal stem cells and 4% hyaluronate (UCB-MSC-HA) will result in reliable cartilage restoration in patients with large, full-thickness cartilage defects and whether any clinical improvements can be maintained up to 5 years postoperatively.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: A randomized controlled phase 3 clinical trial was conducted for 48 weeks, and the participants then underwent extended 5-year observational follow-up. Enrolled were patients with large, full-thickness cartilage defects (International Cartilage Repair Society [ICRS] grade 4) in a single compartment of the knee joint, as confirmed by arthroscopy. The defect was treated either with UCB-MSC-HA implantation through mini-arthrotomy or with microfracture. The primary outcome was proportion of participants who improved by ≥1 grade on the ICRS Macroscopic Cartilage Repair Assessment (blinded evaluation) at 48-week arthroscopy. Secondary outcomes included histologic assessment; changes in pain visual analog scale (VAS) score, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and International Knee Documentation Committee (IKDC) score from baseline; and adverse events.

Results: Among 114 randomized participants (mean age, 55.9 years; 67% female; body mass index, 26.2 kg/m^2), 89 completed the phase 3 clinical trial and 73 were enrolled in the 5-year follow-up study. The mean defect size was 4.9 cm^2 in the UCB-MSC-HA group and 4.0 cm^2 in the microfracture group ($P = .051$). At 48 weeks, improvement by ≥1 ICRS grade was seen in 97.7% of the UCB-MSC-HA group versus 71.7% of the microfracture group ($P = .001$); the overall histologic assessment score was also superior in the UCB-MSC-HA group ($P = .036$). Improvement in VAS pain, WOMAC, and IKDC scores were not significantly different between the groups at 48 weeks, however the clinical results were significantly better in the UCB-MSC-HA group at 3- to 5-year follow-up ($P < .05$). There were no differences between the groups in adverse events.

Conclusion: In older patients with symptomatic, large, full-thickness cartilage defects with or without osteoarthritis, UCB-MSC-HA implantation resulted in improved cartilage grade at second-look arthroscopy and provided more improvement in pain and function up to 5 years compared with microfracture.

Registration: NCT01041001, NCT01626677 (ClinicalTrials.gov identifier).

Keywords: full-thickness cartilage defect; cartilage restoration; mesenchymal stem cells; umbilical cord blood; microfracture

The Orthopaedic Journal of Sports Medicine, 9(1), 2325967120973052

Articular cartilage defects remain a challenging clinical problem. Currently available treatment options are generally

STEM CELLS®
REGENERATIVE MEDICINE

[1]Biomedical Research Institute, R&D Center, MEDIPOST Co., Ltd., Gyeonggi-do, Republic of Korea; [2]Division of Radiation Cancer Research, Korea Institute of Radiological and Medical Sciences, Seoul, Republic of Korea; [3]Leading-edge Research Center for Drug Discovery

Autocrine Action of Thrombospondin-2 Determines the Chondrogenic Differentiation Potential and Suppresses Hypertrophic Maturation of Human Umbilical Cord Blood-derived Mesenchymal Stem Cells

ARTICLE IN PRESS

Osteoarthritis and Cartilage xxx (2016) 1–11

Osteoarthritis and Cartilage

OARSI OSTEOARTHRITIS RESEARCH SOCIETY INTERNATIONAL

Single-stage cell-based cartilage repair in a rabbit model: cell tracking and *in vivo* chondrogenesis of human umbilical cord blood-derived mesenchymal stem cells and hyaluronic acid hydrogel composite[1]

Y.B. Park †, C.W. Ha ‡ § *, J.A. Kim ‡, W.J. Han ‡, J.H. Rhim ‡, H.J. Lee †, K.J. Kim ||, Y.G. Park ||, J.Y. Chung ¶

Published Ahead of Print on September 9, 2016 as 10.5966/sctm.2016-0157.

STEM CELLS TRANSLATIONAL MEDICINE®

TISSUE ENGINEERING AND REGENERATIVE MEDICINE

Cartilage Regeneration in Osteoarthritic Patients by a Composite of Allogeneic Umbilical Cord Blood-Derived Mesenchymal Stem Cells and Hyaluronate Hydrogel: Results From a Clinical Trial for Safety and Proof-of-Concept With 7 Years of Extended Follow-Up

YONG-BEOM PARK,[a] CHUL-WON HA,[b,c,d] CHOONG-HEE LEE,[b] YOUNG CHEOL YOON,[e] YONG-GEUN PARK[f]

[a]Department of Orthopedic Surgery, Chung-Ang University Hospital, Chung-Ang University College of Medicine, Seoul, Republic of Korea; [b]Department of Orthopedic Surgery, [c]Stem Cell & Regenerative Medicine Research Institute, and [e]Department of Radiology, Samsung Medical Center, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea; [d]Department of Health Sciences and Technology, Samsung Advanced Institute of Health Sciences and Technology, Sungkyunkwan University, Seoul, Republic of Korea; [f]Department of Orthopedic Surgery, Jeju National University Hospital

Key Words. Osteoarthritis • Cartilage regeneration • Mesenchymal stem cells • Human umbilical cord blood • Hyaluronic acid

ABSTRACT

Few methods are available to regenerate articular cartilage defects in patients with osteoarthritis. We aimed to assess the safety and efficacy of articular cartilage regeneration by a novel medicinal product composed of allogeneic human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs). Patients with Kellgren-Lawrence grade 3 osteoarthritis and International Cartilage Repair Society (ICRS) grade 4 cartilage defects were enrolled in this clinical trial. The stem cell-based medicinal product (a composite of culture-expanded allogeneic hUCB-MSCs and hyaluronic acid hydrogel [Cartistem]) was applied to the lesion site. Safety was assessed by the World Health Organization common toxicity criteria. The primary efficacy outcome was ICRS cartilage repair assessed by arthroscopy at 12 weeks. The secondary efficacy outcome was visual analog scale (VAS) score for pain on walking. During a 7-year extended follow-up, we evaluated safety, VAS score, International Knee Documentation Committee (IKDC) subjective score, magnetic resonance imaging (MRI) findings, and histological evaluations. Seven participants were enrolled. Maturing repair tissue was observed at the 12-week arthroscopic evaluation. The VAS and IKDC scores were improved at 24 weeks. The improved clinical outcomes were stable over 7 years of follow-up. The histological findings at 1 year showed hyaline-like cartilage. MRI at 3 years showed persistence of the regenerated cartilage. Only five mild to moderate treatment-emergent adverse events were observed. There were no cases of osteogenesis or tumorigenesis over 7 years. The application of this novel stem cell-based medicinal product appears to be safe and effective for the regeneration of durable articular cartilage in osteoarthritic knees. STEM CELLS TRANSLATIONAL MEDICINE *2016;5:1–9*

연골손상 질환단계에 따른 치료요법 및 한계

CARTISTEM

단계	1단계	2단계			3단계
시술법	약물치료	수술적복원술			인공관절 부분/전치환술
		미세천공술 (Microfracture)	자가연골이식술 (OATS)	자가연골세포배양이식술 (ACI)	
대상	• 경미한 연골손상 • RA환자 • 연령제한 없음	• 경미한 손상 • 1단계 효과 부족 시	• 약물치료만으로 효과 부족	• MF등 기존 치료법 효과 부족 시	• 1~2단계 치료가 효과 없을 시
연령 (주연령)	전 연령	15~45세	15~50세	15~50세	60~64세: K&L grade IV 65세 이상:K&L grade III
손상정도 (편측)	전단계적용가능	3~5 cm^2이하	2~4 cm^2	2cm^2~10 cm^2 (유효 size는 5~6cm^2 까지로 보고됨)	4cm^2~20 cm^2 면적과 별개로 손상 심할 경우
장점	• 시술필요 없음 • 적은 비용 • 전연령적용	• 가장 저렴한 시술법 • 침습성이 가장 적음	• 타 수술방법대비 적은 시술비용 • 침습성이 적음	• 자연재생유도가능 • 면역거부 반응 없음	• 높은 만족도 • 확립된 치료법
단점	• 대증적 치료법 • 일정단계이상	• 섬유연골재생경향 (정상의 60%정도) • 3cm^2, 45세 이상 효과 적음	• 적용 가능한 손상범위 제한	• 환자제한(50세 이하) • 2회 시술(연골채취) • 연골상태불량 시 예후 나쁨	• 유효기간(15년) • 재수술 가능

CARTISTEM®

각 단계에서 ICRS Grade IV, K&L Grade I~III, OA, OCD,

2cm^2~9cm^2, No Age Limit 에서 사용가능

Hyaline Like Cartilage 재생

* CARTISTEM®과 HTO(슬관절 내반 변형 5도 이상 급여적용) 병행 수술 시

K&L grade II~III/ ICRS grade IV(medial, lateral) = CARTISTEM + /HTO/DFO 진행 (Varus/Valgus)

K&L grade II~IV/ ICRS grade IV(medial, lateral) = Kissing Lesion은 TKR 진행

■ C-arm의 영향을 최소화 HTO 후 CRTISTEM 수술 진행 권장.

Life-Changing
INNOVATIONS
The Future of Biotechnology, **MEDIPOST**

감사합니다